applied sciences

MDPI

Article

Design, Simulation, and Comparison of Advanced Control Strategies for a 3-Degree-of-Freedom Robot

Claudio Urrea *, John Kern and Víctor Torres

Electrical Engineering Department, Faculty of Engineering, University of Santiago of Chile, Las Sophoras 165, Estación Central, Santiago 9170020, Chile; john.kern@usach.cl (J.K.); victor.torres@usach.cl (V.T.)
* Correspondence: claudio.urrea@usach.cl; Tel.: +56-2-27-183-350

Abstract: This study presents the design, simulation, and comparative analysis of three advanced control strategies applied to a 3-Degree-of-Freedom (DoF) robot manipulator. The controllers investigated are a variant from the Computed Torque Control family, a Proportional–Derivative–Integral with fuzzy logic (PD-PI + fuzzy) controller, and a Model Predictive Control (MPC) scheme. The controller performance is evaluated through the tracking of predefined trajectories in the three-dimensional space. The results are analyzed through XYZ coordinate motion graphs and 3D trajectories. To quantify performance, three error indicators are employed: Residual Mean Square (RMS) with a value of 0.0720 for the Computed Torque Controller, Residual Standard Deviation (RSD), and Index of Agreement (IA). The results demonstrate that the proposed controllers achieve accurate trajectory tracking, with IA values close to unity, demonstrating a high degree of concordance between the desired and executed trajectories.

Keywords: advanced robot control; computed torque control; PD-PI fuzzy control; model predictive control (MPC); nonlinear systems; 3-DoF robot manipulator

check for updates

Citation: Urrea, C.; Kern, J.; Torres, V. Design, Simulation, and Comparison of Advanced Control Strategies for a 3-Degree-of-Freedom Robot. *Appl. Sci.* **2024**, *14*, 11010. https://doi.org/10.3390/app142311010

Academic Editors: Yutaka Ishibashi and Grigorios Beligiannis

Received: 25 September 2024
Revised: 6 November 2024
Accepted: 25 November 2024
Published: 27 November 2024

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

The field of robotics has experienced significant advances in recent decades, driven by progress in control theory and enhanced computational capacity. These advancements have enabled the practical implementation of complex control algorithms, thereby improving the performance of robotic systems in various scenarios [1,2].

Robot manipulators pose unique control challenges due to their nonlinear and coupled nature, specifically regarding the complex interactions between robot joints and the nonlinear dynamics arising from robot configuration and motion. Precise control of these systems is crucial for applications requiring high accuracy and robustness to external disturbances. Therefore, advanced control techniques have proven to be valuable tools for addressing these challenges [3].

Among the most prominent advanced control techniques are the following:

- Computed Torque Control: a family of techniques including variants such as Proportional–Derivative Computed Torque (PDCT) and Proportional–Integral–Derivative Computed Torque (PIDCT).
- Fuzzy Intelligent Control: it employs fuzzy logic to handle system uncertainty and nonlinearity.
- Model Predictive Control (MPC): a technique that utilizes a system model to predict and optimize its future behavior.

This study focuses on the design, simulation, and comparative analysis of these three control strategies applied to a 3-Degree-of-Freedom (DoF) robot manipulator. The study aims to evaluate the performance of each controller in terms of trajectory tracking and robustness to external disturbances. For this purpose, specific trajectories have been

planned and generated for the robot to execute, considering various scenarios and disturbances. Performance evaluation is conducted through graphical analysis and calculation of quantitative performance indices.

Fuzzy logic represents a mathematical approach for modeling imprecise or approximate reasoning and ideas. In fuzzy set theory, an element can belong to a set to varying degrees, allowing partial membership. A fuzzy rule expresses a cause-and-effect relationship through linguistic terms. Its fuzzy nature stems from the use of imprecise and relative adjectives: if X belongs to A, then Y belongs to B, where X and Y are elements and A and B are fuzzy sets [4]. While fuzzy inference systems are universal nonlinear mappers, they still require linear intermediate operations. The following section describes the most common types of fuzzy inference systems [4]:

- Mamdani: Calculates the truth value for each system input. The system employs T-norms, typically using the minimum between inputs or their product. During defuzzification, the membership function is truncated at the T-norm value level.
- Takagi-Sugeno: This system enhances Mamdani's approach by reducing computational processing. Instead of output membership functions, it uses concrete linear combinations of input values. The final output is computed as a weighted average between the linear output combinations and T-norm values.
- Tsukamoto: Similarly to Mamdani, this system uses injective output membership functions to obtain the pre-image of the T-norm value.

The authors in [5] present novel fuzzy metrics following George and Veeramani's approach. They classify these examples according to their construction and established fuzzy metrics, demonstrating their application to color image processing with potential extensions to other engineering problems.

Equations (1)–(3) present well-known generic fuzzy metrics [5].

$$M(x,y,t) = \left(\frac{(min\{f(x),f(y)\})^{\alpha} + g(t)}{(max\{f(x),f(y)\})^{\alpha} + g(t)} \right)^{\beta} \quad (1)$$

$$M(x,y,t) = \frac{g(t)}{g(t) + m * d(x,y)} \quad (2)$$

$$M(x,y,t) = e^{(-d(x,y)/g(t))} \quad (3)$$

Subsequently, the authors in [6] implement a linear MPC scheme for a Shunk PowerCube robot model, achieving reduced computational load and semi-automatic trajectory execution. The system integrates camera-based feedback to demonstrate vision-based control effects and improve control performance. Their experimental validation includes two experiments, highlighting the advantages and limitations of their vision-based time-varying linear MPC scheme implemented on standard computing hardware.

In [7], the authors developed an MPC with vision-based servo control to achieve high-precision control for hand-like object manipulation, addressing the complexities of modeling grippers, objects, and contacts. Their control scheme demonstrates superior performance compared to conventional vision-based servo control.

In [8], researchers present a Learning-Based Nonlinear Model Predictive Control (LB-NMPC) algorithm designed to achieve high-performance path tracking in challenging terrains through learning. The system integrates a simple vehicle model with added perturbation, where perturbations are modeled as a Gaussian process based on system state, input, and other relevant variables. The system achieves mobile localization through an integrated vision-based navigation and mapping system, enabling operation in large-scale environments without a Global Positioning System (GPS). Experimental validation spans over 3 km of traversal across three distinct robotic platforms with masses ranging from 50 to 600 kg at speeds between 0.35 and 1.2 m/s. Results confirm that the controller can initialize from a generic vehicle model and progressively reduce vehicle-specific and trajectory-specific path-tracking errors through experience.

In [9], researchers developed and simulated Fuzzy Logic Joint Controllers (FLJC) for a 6-DoF robot arm integrated with an artificial vision system for color discrimination. The system's performance was evaluated by comparing computational load, speed, and precision through displacement measurements in XYZ Cartesian axes across two and three dimensions. Results indicate an absolute error below 2 cm for gripper coordinates, while the artificial vision system maintains at least 95% accuracy in color discrimination for objects stacked in both the first and second levels.

In [10], the research focuses on the design process, kinematic modeling, and parameterization of a 4-DoF SCARA-type robotic arm specifically engineered for agricultural environments. Current research continues on robotic arms and manipulation systems since commercially available solutions are not optimized for agriculture's dusty, dirty, and humid conditions. The project aims to develop a robotic arm capable of operating in adverse agricultural conditions, including rain, wind, dust, mud, and vibration, featuring an optimized number of DoF. This optimization enables adaptability to real agricultural conditions through specially designed joint components and lightweight links with a fast and simplified control unit. Initial results demonstrate workspace coverage of 11.4 m^2 for maximum frontal access and 2.9 m and 2.4 m for lateral access, confirming sufficient surface coverage for agricultural applications.

Based on the aforementioned works, research predominantly employs simulations, implementing various control strategies that incorporate advanced control theory fundamentals, notably combining fuzzy logic and predictive control with emerging technologies like vision sensors. There exists a growing need to develop proprietary technologies aligned with the socio-economic requirements of each region, particularly in agriculture, medicine, and mining.

In [11], it is indicated that the robotic manipulator has become an integral component of modern industrial automation. In their study, a model of a robotic arm with 6 DoF is presented in which two modern control strategies, H∞ and MPC, are applied. The designed control laws are subjected to various step and ramp inputs; their effectiveness is evaluated on transient parameters such as settling time, overshoot, and steady-state error. For the MPC controller, it is observed that the system response is related to the control window size; its performance improves when this size is increased. In their results, optimal values of np = 100 and nc = 20 were obtained. For the H∞ controller, the weighting functions have a significant effect on the performance of the designed control law, where these functions are used to trace the ramp, step, and sinusoidal responses of the system. From the graphs, all joints of the robotic arm showed similar behavior, with a delay in the controller response for the reference trajectories. The comparative analysis of the performance achieved by both controllers reveals that the MPC controller outperforms the H∞. In the future, it is planned to implement both control strategies on a real robotic platform. It is also proposed to investigate the performance of control when the robotic arm is subject to disturbances and uncertainties, as well as an application-oriented study to explore practical avenues for research.

In [12], the authors present a redundant planar manipulator with six degrees of freedom (6-DoF) that features an innovative structural configuration including rotary and prismatic joints, to which three control strategies are applied for monitoring and regulating the joint trajectory tracking problem of the robot under variable loads. The selected control laws were MPC and fuzzy logic controller (FLC), subsequently comparing them with the classical Proportional–Integral–Derivative (PID) controller. The simulation models the behavior of the redundant planar manipulator in a virtual environment prior to its physical construction in order to conduct performance tests of its controllers and anticipate possible damages/failures in the system mechanics before implementing control strategies on a real robot. The effectiveness of these three control strategies is validated through qualitative comparisons (trajectory tracking performance graphs) and quantitative comparisons (the performance indicator of the Common Mode Rejection Ratio (CMRR) and joint error indices, such as the Residual Mean Square (RMS), the Residual Standard Deviation (RSD), and the

Index of Agreement (IA)). It has been demonstrated that the conventional PID controller performs well in linear systems; however, it is challenging to tune the controller parameters, as most industrial plants are represented by complex dynamic systems of higher order that do not include linearities or time delays. FLC control has been one of the most successful intelligent control techniques, and its main advantage is that it does not depend on the dynamics of the robotic manipulator. Furthermore, the fuzzy controller enhances the system's performance by minimizing disturbances and uncertainties.

The MPC technique has also become a powerful tool for controlling both linear and nonlinear systems. In recent decades, the trajectory tracking problem has been widely investigated with the aim of reducing the tracking error to zero, where robotic manipulators often suffer from various uncertainties, such as unstructured dynamics, external disturbances, unknown load variations, and nonlinear friction, which degrade the system's performance. Moreover, the adaptive control strategy is mentioned, which requires a constant unknown parameter; this condition is only met if the load of a robotic manipulator is fixed, which does not occur in practice, as a robotic manipulator often needs to pick up and drop specific loads repeatedly, leading to jumps in the values of the robot's unknown parameters. Regarding the results obtained, it can be mentioned that the PID controller is feasible in redundant robots; however, its performance was not good when the applied variable load exhibited linear characteristics. By adopting the fuzzy method, the uncertainties and disturbances of the system were managed, and control of disturbance attenuation was achieved. It is also indicated that the control signal from the FLC was smooth and did not present significant instabilities during the transient regime, except at the moment when the robot's joints exited their inertia. When applying variable loads to the robot, its recovery was adequate, as it did not exhibit overshoots or jerky movements in its joints compared to the PID controllers and the MPC-based controller; it was confirmed that the obtained RMS/RSD values were close to zero and those of the IA were close to one when using the FLC. Moreover, the high CMRR values of the FLC demonstrated that the compensation performed by these controllers, when variable loads were applied, was much more effective than that of the PID and MPC-based controllers. Thus, after comparing the three control methodologies, the FLCs proved to be the most suitable and efficient, both under normal operating conditions and under the influence of variable loads applied to the robot's joints.

The authors [13] indicate that the SCARA robot is one of the most established robots in the industry due to its high speed and precision, with the controller being key to ensuring its performance. This study presents three sections dedicated to some advanced controllers, as well as the Internet of Things (IoT) and Artificial Intelligence (AI) applying neural networks, particle swarm optimization, and genetic algorithms.

For the controllers, the PID controller is described as a closed-loop control system with gains whose magnitude affects the transient response. Among the characteristics of the PID controller are its stability in a system that behaves linearly or nearly linearly and its difficulty in nonlinear systems; it also has insufficient sensitivity to counteract high-speed and precision operational requirements. To overcome the linear response of the PID controller, many have adopted approaches to actively change the PID gain according to the operating condition; one of these is the use of fuzzy logic and AI algorithms.

Another type of control is fuzzy logic, which derives from mathematical logic to manipulate the signal between intervals of 0 and 1. Its operation is based on input, inference engine, and output, which will be used to control the robotic arm according to the PID controller gains. This overcomes the fixed gains that require prior tuning and the lack of adaptability to the nonlinear behavior of the SCARA robot, allowing for the alleviation of the PID controller limitation by adding fuzzy logic to improve the smoothness and stability of the robot.

There is sliding mode control, which is a nonlinear dynamic control system; its purpose is to control state variables of the system towards a designated surface known as the sliding surface and to ensure that the trajectory of the state variable remains on this designated surface. It has characteristics that provide high precision, control parametric

uncertainties, and external disturbances, but its disadvantage is that it can cause high-frequency oscillations in the control signal in a discontinuous control system.

Adaptive control is also one of the advanced controllers that can handle nonlinear dynamic systems. Considering that the plant for the SCARA robot is modeled as a linear time-invariant plant in continuous time, the reference model can be represented using the general state-space model according to the adaptation mechanism, which is the adaptive control law; this will continuously adjust the tunable parameter in the controller according to the operating environment based on the reference model.

Robust control is a type of controller designed to improve the controller's performance over a specified range. The idea of robust control was proposed by Evan Anderson and Paulm Dupuis. The authors, in their conclusions, indicate that the selection of the controller must align with the robot's application due to its respective advantages and disadvantages. If the application involves linear behavior, PID control and fuzzy control are the correct choice, as both control systems rely on fixed gains; PID and FLC are suitable for fixed-load robots that do not have much uncertainty. On the other hand, sliding mode, adaptive, and robust control are suitable for handling high-uncertainty situations. In relation to AI technology, it is noted that it can help the advanced controller achieve better performance than the SCARA robot; however, such powerful AI requires a computer with high hardware specifications. Moreover, IoT can create a more user-friendly environment, as it allows for controlling the SCARA robot with a smartphone.

- Findings: a study of three advanced controllers for application in a 3-DoF robot is provided.
- Practical implications: the system utilizes the dynamic model of a Scorbot robot, considering 3-DoF.
- Originality/value: the simulation method of three controllers is proposed through performance comparisons with trajectory tracking graphs and performance indicators.

Our study primarily aims to obtain the dynamic model and parameterization of a 3-DoF Scorbot-type robotic arm to enable the design, simulation, and comparison of three advanced controllers meeting speed, practicality, adaptability, and scalability requirements. The selected robotic system resembles the Scorbot robot intended for mining industry operations, with simulation providing an ideal method to evaluate potential real operation scenarios under complex conditions. The approach remains applicable to any robotic arm. While practical implementation faces time and resource limitations, simulation offers significant advantages in safety assurance through preliminary controller evaluation. The research aims to provide a valuable guide for researchers and professionals, highlighting the significance of controllers in robotic arm applications.

2. Literature Review

This section presents the state-of-the-art research on a 3-DoF robot and advanced control techniques.

2.1. Advances in Computed Torque Control

References [1–3] provide foundational literature on robotic systems. The first two works comprehensively cover theoretical and practical aspects of manipulators, emphasizing kinematics, dynamics, trajectories, and both linear and nonlinear control, along with programming languages. The third reference specifically addresses advanced robotic manipulator control, beginning with the fundamentals and development of dynamic equations for a 2-DoF robot. It then introduces Computed Torque Control as a fundamental control method, followed by simulation examples, advanced control definitions, and applications.

In [14], researchers detail the design and implementation of a self-developed 5-DoF. They implement a Computed Torque Control in joint space for trajectory following. The study employs direct and inverse kinematics for robot modeling and utilizes dynamic equations to verify trajectory generation experimentally. The authors suggest exploring alternative nonlinear controllers in future work.

Research presented in [15] explores redundant robots, which possess more DoF than required for specific motion tasks. This redundancy enables additional capabilities, such as joint limit and singularity avoidance. The authors propose and validate a two-loop motion control scheme through experiments on a laboratory-developed planar redundant arm.

The foundational literature establishes the theoretical framework necessary for analyzing robot manipulator operations across medical, industrial, and scientific applications. The specialized research on modern controllers provides comprehensive insights into robot manipulator control, complemented by simulation methodologies. Further studies demonstrate the effectiveness of Computed Torque Control in nonlinear systems, particularly when integrated with dynamic modeling and kinematic redundancy, enabling additional capabilities such as obstacle avoidance. The present research investigates a 3-DoF robot system, incorporating dynamic modeling and disturbance considerations to achieve precise geometric trajectory execution through advanced Computed Torque Control, emphasizing smooth motion characteristics.

2.2. Developments in Fuzzy Control and Hybrid Systems

Research by [16] demonstrates the industrial applications of robot arms using a dual-control approach: a fuzzy controller for effective object detection and a Proportional–Integral–Derivative (PID) controller for motion stability. Their experimental results confirm that proper K_p and K_d parameter configuration ensures stable transient and steady-state behavior in "Pick and Place" operations.

In [17], researchers address the complexities of nonlinear system control, particularly in robot arm modeling. They categorize control strategies into classical methods, based on conventional control theory, and non-classical methods, utilizing intelligent system theory (fuzzy, neural networks, and neuro-fuzzy). The study highlights the challenges in modeling dynamic parameters such as friction and inertia, proposing a neuro-fuzzy hybrid control evaluated through Root Mean Square error measurements.

The study presented in [18] details the modeling, simulation, and control of a 2-DoF robot arm through direct and inverse kinematics. The researchers derive the robot's dynamic equation using the Euler–Lagrange method and implement their simulation using MATLAB/Simulink's Simscape Multibody and Robotics System Toolbox with a fuzzy logic controller.

These research findings demonstrate that fuzzy intelligent control effectively addresses system complexity in robotic control through two key mechanisms: sensor input processing and management of challenging dynamic parameters such as friction and inertia, complemented by PID control integration. Building upon these foundations, this study proposes a hybrid PD-PI fuzzy control architecture designed to enhance both stability and motion smoothness. In this context, hybrid systems represent control architectures that synergistically combine multiple control strategies—such as PID and fuzzy control—to capitalize on their respective advantages.

2.3. Applications of Predictive Control in Robotics

In [19], experts present a comprehensive analysis of linear MPC and Nonlinear Predictive Control (NMPC). The study emphasizes that NMPC requires addressing a nonlinear non-convex problem (NPL), presenting computational challenges in cost and reliability. The authors detail Sequential Quadratic Programming (SQP) methods, which extend Newton-type methods to achieve Karush–Kuhn–Tucker (KKT) condition convergence. This approach must ensure rapid convergence while managing ill-conditioning and extreme nonlinearities.

Research presented in [20] acknowledges the highly nonlinear nature of robot models and proposes multiple control approaches, including fuzzy and neural control. The study develops a PID-NMPC-based robot with an optimal time trajectory using convex optimization. Their methodology employs a two-step design: initial system linearization followed by MPC application, comparing results with optimal Linear Quadratic (LQ) control.

In [21], researchers explore predictive control as an optimal method for robot planning and obstacle avoidance. They introduce a model-free predictive controller capable of operation without prior kinematic and dynamic knowledge. The controller utilizes MPC to predict system states and generate optimal control signals over a defined prediction horizon.

MPC is used in both linear and nonlinear systems, with the latter applied to a robot manipulator, enabling its use in robot path planning and obstacle avoidance. It should be noted that MPC requires an accurate model of the system, making it essential to use the robot's dynamic model with the greatest detail. This document presents an MPC applied to a 3-DoF Scorbot robot.

2.4. Optimization and Comparative Analysis of Controllers

Recent research by [22] presents a comparison between two optimization techniques, Artificial Bee Colony (ABC) and Artificial Bee Colony with Multielite Guidance (ABCMG), for optimizing a PID controller on a 3-DoF robot. Their results demonstrate that the MGABC algorithm outperforms the traditional ABC based on multiple performance indicators. Furthermore, the optimized system exhibits robustness under various disturbance conditions, suggesting that the proposed techniques and their objective function are viable for PID controller optimization in robotic applications.

Research presented in [23] addresses the nonlinear behavior challenges in robot manipulators. The study implements and evaluates five controllers on a 3-DoF robot: predictive, hyperbolic sinusoidal, sliding mode, a hybrid predictive + hyperbolic cosine, and adaptive. The researchers established a communication interface between MATLAB/Simulink and the robot to generate trajectories for analyzing the dynamic performance using Residual Mean Square indicators.

In [24], researchers design and assess four controllers: two classical (Computed Torque and PID) and two advanced (fuzzy logic and predictive) implementations for a SCORBOT-ER V plus robot model. Their simulation incorporates gravity compensation, inverse motor models, and belt movement effects in each joint. Controller evaluation is based on trajectory curves, joint errors, and three performance indices.

Notable advances in the field include novel algorithms and controllers recently developed for robotics. Contemporary approaches such as Artificial Bee Colony, Genetic, Particle Swarm, Harmony Search, and Differential Evolution algorithms offer new opportunities for control applications in robot manipulators. The latter two studies provide comprehensive analyses through controller comparisons and detailed supporting annexes. Building upon these works, this research presents three advanced controllers, alongside a PID control for comparison, including the simulation of obstacle avoidance along the robot's trajectory.

2.5. Advanced Control Techniques

Computed Torque Control uses the following equations, which represent the fundamental control law applied to a 3-DoF robot:

- PD Computed Torque Control Law, which can be represented by Equation (4)

$$\tau = M(q)\left(\ddot{q}_d + K_p e + K_d \dot{e}\right) + C\left(q, \ddot{q}\right) + G(q) + F(\dot{q}) \quad (4)$$

- PID Control Law, which can be represented by Equation (5)

$$\tau = K_p e + K_d \dot{e} + K_i \int_0^t e dt \quad (5)$$

For the case of intelligent control, the fuzzy control law can be mentioned. There is the possibility of combining some controls, for example, the PID control plus the fuzzy control, obtaining a hybrid type of control.

Predictive control [19] is generally designed based on an objective function that is sought to be minimized, for example, optimizing the robot's trajectory to avoid obstacles that appear in its path. As discussed in the state of the art, for the development of the

work, the modeling of a 3-DoF robot, the Scorbot model, and the design of the Computed Torque family controllers, PD-PI + fuzzy, and Model Predictive Control are considered, structured, and developed based on the theoretical and technical information indicated in the bibliography.

The structure of this article is as follows: Section 2 presents a review of the state of the art in robot manipulator control. Section 3 describes the robotic system used in the study. Section 4 details the design of the proposed controllers. Section 5 presents the simulation results. Section 6 discusses the main findings. Section 7 addresses the conclusions of the research, and the projections for future work are raised in Section 8.

3. Description of the System Under Study

This study utilized MatLab® R2024a software [25] on an Asus Vivobook laptop with an Intel 2.5 GHz processor and 16 GB RAM. The robot selected to validate the performance of the proposed control strategy is the Scorbot robot. This is a robot that has been used in numerous applications due to its characteristics and capabilities.

To simulate the robot's motion, the dynamic equation model associated with the particular model must generally be used, which is obtained with computer programs and written development. Figure 1 shows the 3-DoF Scorbot robot [24–27].

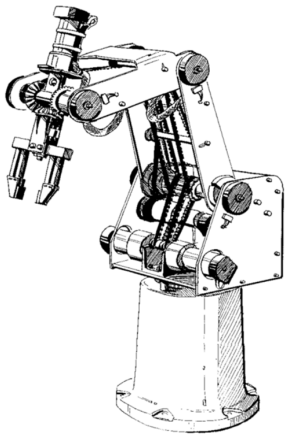


Figure 1. Scorbot robot.

3.1. Scorbot Robot Structure

The Scorbot robot, described in Figure 2, has a base that is normally fixed to a working surface for the first DoF (base), extending its work area with the two remaining links corresponding to the second (shoulder) and third (elbow) DoF, adding the incorporated end-effector with good accuracy in its displacement. However, it can be attached to a sliding base, allowing a range of work with greater spatial coverage.

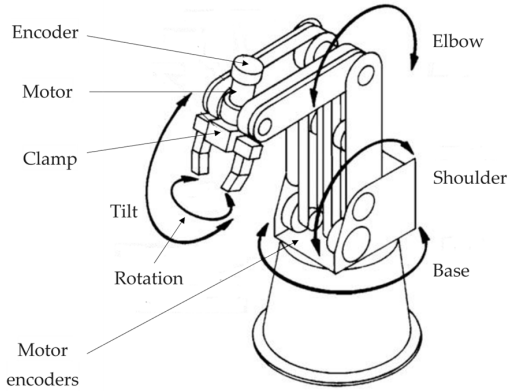


Figure 2. Scorbot robot morphology.

Among its structural and technical characteristics, the number of axes, end-effectors, and actuators, among others, which are listed in Table 1, can be mentioned.

Table 1. Scorbot robot specifications [26].

SCORBOT-ER V Plus Specifications	
Mechanical structure Number of axes	Vertical articulated Five axes plus servo gripper (three axes are considered)
Axis Movement Axis 1: base rotation (first DoF) Axis 2: shoulder rotation (second DoF) Axis 3: elbow rotation (third DoF) Axis 4: wrist pitch (end effector) Axis 5: wrist roll (end effector)	 310° +130°/−35° ±130° ±130° Unlimited (mechanically); ±570° (electrically)
Maximum operating radius	610 mm (24.4″)
End effector	DC servo gripper with optical encoder, parallel finger motion; Measurement of object's size/gripping force by means of gripper sensors and software.
Maximum gripper opening	75 mm (3″) without rubber pads 65 mm (2.6″) with rubber pads
Hard home	Fixed position on each axis, found by means of microswitches
Feedback	Optical encoder on each axis
Actuators	12V DC servo motors
Motor capacity (axes 1–6)	15 oz. in Peak Torque (stall) 70 W Power for Peak Torque
Gear ratios	Motors 1, 2, 3: 127.1:1 Motors 4, 5: 65.5:1 Motor 6 (gripper) 19.5:1
Transmission	Gears, timing belts, and lead screw
Maximum payload	1 kg (2.2 lb.), including gripper
Position repeatability	±0.5 mm (0.02″) at TCP (tip of gripper)
Weight	11.5 kg (25 lb)
Maximum path velocity	600 mm/s (23.6″/s)
Ambient operating temperature	2°–40 °C (36°–104°F)

3.2. Kinematic Model of the Scorbot Robot

The kinematics of the robot manipulator are obtained using the Denavit–Hartenberg (DH) algorithm, described in Table 2 [27].

Table 2. DH parameters of the Scorbot robot.

Link	$\boldsymbol{\theta_i}$	$\boldsymbol{d_i}$	$\boldsymbol{a_i}$	$\boldsymbol{\alpha_i}$
1	θ_1	l_1	0	90°
2	θ_2	0	l_2	0
3	θ_3	0	l_3	0

Using the parameters from the previous table, the transformation matrix of each element of the kinematic chain is obtained according to (6) to (14). For simplicity, the following definitions are used:

$$C_i = \cos\theta_i \quad (6)$$

$$S_i = \sin\theta_i \tag{7}$$

$${}^0A_1 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

$${}^1A_2 = \begin{bmatrix} C_2 & -S_2 & 0 & l_2C_2 \\ S_2 & C_2 & 0 & l_2C_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9}$$

$${}^2A_3 = \begin{bmatrix} C_3 & -S_3 & 0 & l_3C_3 \\ S_3 & C_3 & 0 & l_3S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

Using the previous matrices, the homogeneous transformation matrix of the complete kinematic chain can be obtained.

$$T\begin{bmatrix} C_1C_2C_3 - C_1S_2S_3 & -C_1C_2S_3 - C_1S_2C_3 & S_1 & l_3C_1C_2C_3 - l_3C_1S_2S_3 + l_2C_1 \\ C_2C_3 - S_1S_2S_3 & -S_1C_2S_3 - S_1S_2\,C_3 & -C_1 & l_3S_3C_2C_3 - l_3S_1\,S_2\,S_3 + l_3S_1C_2 \\ S_2C_3 + C_2S_3 & -S_2S_3 + C_2C_3 & 0 & l_3S_2C_3 + l_3C_2S_3 + l_2S_2 + l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{11}$$

The last column of matrix T in Equation (11) corresponds to the position equations of the end-effector.

$$P_x = l_3C_1C_2C_3 - l_3C_1S_2S_3 + l_2C_1C_2 \tag{12}$$

$$P_y = l_3S_3C_2C_3 - l_3S_1S_2S_3 + l_3S_1C_2 \tag{13}$$

$$P_z = l_3S_2C_3 + l_3C_2S_3 + l_2S_2 + l_1 \tag{14}$$

Equations (12)–(14) form the direct kinematic model of the Scorbot robot.

3.3. Dynamic Model of the Scorbot Robot

The dynamic model of the Scorbot robot is obtained through the Euler–Lagrange equations, using (15) and (16) to form (17) and applying (18) to form (19).

$$K_i(q,\dot{q}) = \frac{1}{2}m_iv^2 + \frac{1}{2}I_i\ddot{\theta^2} \tag{15}$$

$$U_i(q) = m_igh \tag{16}$$

$$L(q,\dot{q}) = K_i(q,\dot{q}) - U_i(q) \tag{17}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_i}\right) - \frac{\partial L}{\partial \theta_i} = \tau \tag{18}$$

By adding the terms $F(q) + \tau_d$, Equation (19) is derived.

$$\tau = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + F(q) + \tau_d \tag{19}$$

The Scorbot robot shown in Figure 3 employs three degrees of freedom, allowing it to determine its position in space (x, y, z) [24].

Figure 4 presents the diagram of the robot under study. To model its kinematics and dynamics, the coordinate axes and centroid systems consider q_1, q_2, q_3 and I_1, I_2, I_3 as generalized coordinates and lengths of the first, second, and third links, respectively. Furthermore, I_{c1}, I_{c2}, I_{c3} represent the lengths of the first, second, and third links from the origin to the centroid [24].

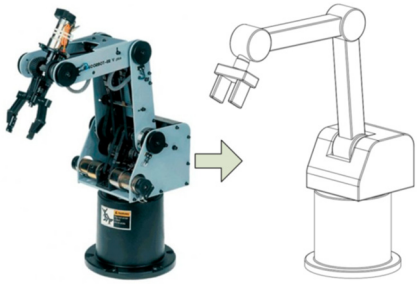
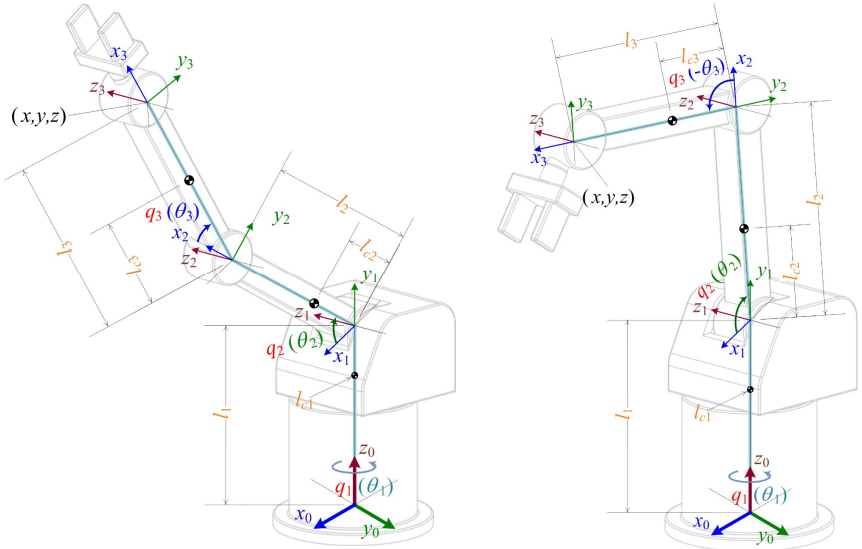


Figure 3. Robot SCORBOT-ER V plus and its diagram.



(**a**) (**b**)

Figure 4. Diagram of the SCORBOT-ER V plus robot in two positions, considering the coordinate axes and centroid systems. (**a**) Elbow down. (**b**) Elbow up.

This results in the matrix M (20) and the vectors C (30), G (34), and F (38).

$$M = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix} \tag{20}$$

$$M_{11} = (l_{c3}\cos(\theta_2 + \theta_3) + l_2\cos(\theta_2))^2 m_3 + I_{1zz} + I_{2zz} + I_{3zz} + \left(\frac{\left(l_{c2}{}^2\cos(2\theta_2)\right)}{2} + \frac{l_{c2}{}^2}{2}\right) m_2 \tag{21}$$

$$M_{12} = I_{2zz} + I_{3zz} \tag{22}$$

$$M_{13} = I_{3zz} \tag{23}$$

$$M_{21} = I_{2zz} + I_{3zz} \tag{24}$$

$$M_{22} = \left(l_2{}^2 + 2\cos(\theta_3) l_2 l_{c3} + l_{c3}{}^2\right) m_3 + m_2 l_{c2}{}^2 + I_{2zz} + I_{3zz} \tag{25}$$

$$M_{23} = \left(l_{c3}{}^2 + l_2\cos(\theta_3) l_{c3}\right) m_3 + I_{3zz} \tag{26}$$

$$M_{31} = I_{3zz} \tag{27}$$

$$M_{32} = \left(l_{c3}{}^2 + l_2\cos(\theta_3) l_{c3}\right) m_3 + I_{3zz} \tag{28}$$

$$M_{33} = l_{c3}{}^2 m_3 + I_{3zz} \tag{29}$$

$$C = \begin{bmatrix} C_{11} \\ C_{21} \\ C_{31} \end{bmatrix} \tag{30}$$

$$\begin{aligned} C_{11} = -dq_1 dq_2 \Big(& m_2 \sin(2\theta_2) {l_{c2}}^2 + 2m_3 (l_{c3}\cos(\theta_2+\theta_3) + l_2\cos(\theta_2))(l_{c3}\sin(\theta_2+\theta_3) + l_2\sin(\theta_2)) \Big) \\ & -2dq_1 dq_3 l_{c3} m_3 \sin(\theta_2+\theta_3)(l_{c3}\cos(\theta_2+\theta_3) + l_2\cos(\theta_2)) \end{aligned} \tag{31}$$

$$\begin{aligned} C_{21} = \Big(\Big(\tfrac{m_2 \sin(2\theta_2) {l_{c2}}^2}{2} \Big) + m_3 (l_{c3}\cos(\theta_2+\theta_3) + 12\cos(\theta_2))(l_{c3}\sin(\theta_2+\theta_3) + l_2\sin(\theta_2)) \Big) {d\theta_1}^2 \\ -l_2 l_{c3} m_3 \sin(\theta_3) {d\theta_3}^2 - 2dq_2 l_2 l_{c3} m_3 \sin(\theta_3) d\theta_3 \end{aligned} \tag{32}$$

$$C_{31} = l_{c3} m_3 \sin(\theta_2+\theta_3)(l_{c3}\cos(\theta_2+\theta_3) + l_2\cos(\theta_2)) {d\theta_1}^2 + l_2 l_{c3} m_3 \sin(\theta_3) {d\theta_2}^2 \tag{33}$$

$$G = \begin{bmatrix} G_{11} \\ G_{21} \\ G_{31} \end{bmatrix} \tag{34}$$

$$G_{11} = 0 \tag{35}$$

$$G_{21} = g_z (m_3 (l_{c3}\cos(\theta_2+\theta_3) + l_2\cos(\theta_2)) + l_{c2} m_2 \cos(\theta_2)) \tag{36}$$

$$G_{31} = g_z l_{c3} m_3 \cos(\theta_2+\theta_3) \tag{37}$$

$$F = \begin{bmatrix} F_{11} & F_{21} & F_{31} \end{bmatrix}^T \tag{38}$$

where

m_1, m_2, and m_3: represent the masses of links one, two, and three, respectively.

I_{1zz}, I_{2zz}, and I_{3zz}: represent the moments of inertia of the first, second, and third links with respect to the first z-axis of their respective joints to facilitate their modeling [24].

Nonlinear control in a robot uses the dynamic equation described in (19) [1,2], where

τ: vector of torques and forces with dimension $(nx1)$.
M: inertia matrix with dimension (nxn).
C: vector of Coriolis and centrifugal forces with dimension $(nx1)$.
G: gravity vector with dimension $(nx1)$.
F: friction vector with dimension $(nx1)$.
τ_d: externals torques and forces with dimension $(nx1)$ *in* [N·m].
n: number of DoF.
q: represents the position term.
$\dot{q}$: represents the velocity term.
$\ddot{q}$: represents the acceleration term.

Next, Table 3 presents the Scorbot model with some of its real technical specifications for modeling.

Table 3. Scorbot model's technical specifications [24,26].

Symbol	Link 1	Link 2	Link 3	Unit
l	0.32	0.22	0.22	[m]
lc	0	0.052	0.1376	[m]
m	7.1402	2.2483	1.957	[kg]
Izz	0.04624	0.02595	0.03616	[kg·m^2]
Fv	0.025	0.025	0.025	[N·m/rad]
Feca	0.05	0.05	0.05	[N·m]
Fecb	0.05	0.05	0.05	[N·m]

3.4. Choice of Robot Under Study

One of the central challenges in controlling robot manipulators is ensuring that the end-effector follows a desired trajectory, reaching a specific target position. The Scorbot robot, with its suitable link lengths and joint rotation degrees, offers an optimal workspace

for such studies. Its structural and technical characteristics make it an ideal candidate for the examination and analysis of various control algorithms. Additionally, many large-scale machines used in the mining industry, such as excavators, drills, and rock breakers, share similar structural designs with the Scorbot. This makes the analysis of the Scorbot relevant to the future automation of heavy machinery used in mining operations like comminution processes.

4. Design of Control Strategies

This section outlines the theoretical foundation of the proposed control strategy.

4.1. Dynamic Control

Robots require high-performance controls to ensure speed and precision. Advanced materials used in manufacturing allow robots to improve movement significantly. However, the dynamic characteristics of robots, such as inertia and friction, often unknown, complicate the task of aligning the actual trajectory with the desired one. Dynamic control aims to ensure that the robot follows the desired trajectory by analyzing its dynamics. Typically, this control is performed in the joint space to manage joint trajectories. If the robot interacts with the environment, Cartesian (task) space control is preferable, controlling the end-effector's trajectory.

Advanced control techniques include the following:

- Adaptive Control: adapts to changes in the robot's dynamics during operation.
- Fuzzy Control: addresses difficulties in creating mathematical models.
- Predictive Control: uses process models to generate control signals, typically requiring high computational effort but useful for multi-input, multi-output systems.

A basic problem in robot control is to make it follow a planned trajectory. There are many issues associated with path planning, such as obstacle avoidance and ensuring that the planned trajectory does not exceed the voltage and torque limitations of the robot's actuators. Computed Torque Control is a classic and fundamental control technique that produces a family of easily understandable control schemes that work well in practice. In addition, the robot is a rigid and well-known system, which makes the design of the controllers based on an established model feasible. Simulating the robot controllers on a computer allows for verifying the effectiveness of the control scheme for any proposal before actual implementation, since in most practical situations, the controllers are implemented in microprocessors associated with the system's electronics. One way to classify the robot control schemes is to divide them into "similar to Computed Torque Control" or "similar to Non-Computed Torque". Computed Torque type controls are mentioned in various bibliographies on robust control, adaptive control, learning control, and others [1]. Table 4 shows a comparison of the controllers mentioned above.

Table 4. Controller comparison [13,28,29].

Controller	Weaknesses	Strengths	Complementary Characteristics
Computed Torque (Adaptive Control)	None observed.	Guarantees global asymptotic stability and, in practical terms, calculates the exact amount of energy required by servomotors to execute a planned trajectory. Cancels robot dynamics in closed-loop and reduces the computation of calculations to solve the system in state variables during numerical integration.	Applicable to any robot with n DoF.

Table 4. *Cont.*

Controller	Weaknesses	Strengths	Complementary Characteristics
Fuzzy Control	Cannot eliminate steady-state error.	Provides better performance in terms of settling time, response time, overshoot, and robustness. Stable control. High precision. Able to adapt to different operating conditions.	Applicable to other robot types.
Model Predictive Control	Requires knowledge of a sufficiently accurate dynamic system model. Requires an optimization algorithm, thus can only be implemented by computer. Requires high computational cost, making it difficult to apply to fast systems. Complex consideration of uncertainties.	Executes constraints and can be used with Multiple-Input Multiple-Output (MIMO) systems. Time-domain formulation, flexible and open. Allows handling of linear and nonlinear systems, Single-Variable and Multi-Variable using the same controller formulation. Control law responds to optimal criteria. Allows incorporation of constraints in controller synthesis.	Applicable to other robot types.
PID	Insufficient for the nonlinear control system. Low precision. Unable to adapt to different operating conditions.	Stable control. Easy to apply.	Used at industrial level.

4.2. Control Scheme

The control scheme for a nonlinear system, in this case for an application using a robot, is presented below and shown in Figure 5 [3,14,30].

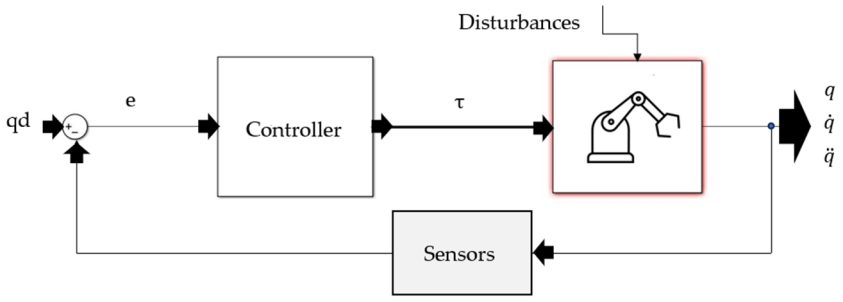


Figure 5. Control scheme.

Similarly to other feedback control systems, the following terms can be distinguished:

- q_d: reference signal called the desired trajectory.
- q, q' y q'': signals to be controlled called the actual trajectories associated with motion, velocity, and acceleration.
- e: error produced by the difference between q_d and q.
- Controller: generally, includes parts of the robot's dynamic equation.
- System model: corresponds to the mathematical model of the robot, represented as a matrix and vectors.
- Disturbance: external variables that cause changes in the robot's motion.
- Sensors: elements in the feedback loop that allow the identification of the variables to be controlled.

4.3. Design of the 3-DoF Robot

To observe and analyze the robot's motion, it is necessary to use the associated dynamic model. Currently, the MATLAB/Simulink V2024 simulation software offers the Simscape

Toolbox, which allows for designing a robot using blocks to visualize its movements. For the design of the 3-DoF robot, the Simscape Multibody and Robotics System Toolboxes of MATLAB/Simulink are used.

Scorbot Robot Model

Similarly, Figure 6 shows a schematic of a 3-DoF robot similar to the Scorbot model.

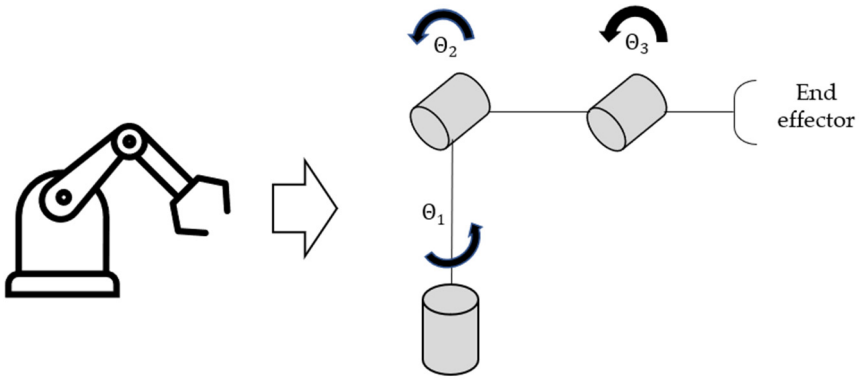


Figure 6. 3-DoF Scorbot robot.

Figure 7 shows a schematic of the Scorbot robot considering the length of its links.

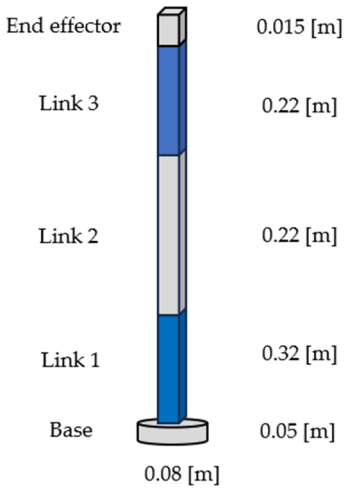


Figure 7. Scorbot robot considering the link lengths.

4.4. Controllers

The methodology for controller design, testing, and validation is summarized as follows:

4.4.1. Computed Torque Family Controller

The following Computed Torque Control family is presented:

Proportional–Derivative Computed Torque Control

Computed Torque Control, a nonlinear control method based on the dynamic model, is particularly suited for robot control. This approach, which represents a special application of feedback linearization in nonlinear systems, has become increasingly relevant in modern system theory. Its effectiveness is optimal when robot arm parameters are accurately known. A key challenge in robot control involves ensuring the manipulator accurately follows a previously planned trajectory. Precise positioning is prerequisite to any useful task. High-performance operation in uncertain environments requires modern design

techniques. Additionally, consideration must be given to robot movement in free space, where contact with the environment generates external forces that may affect joint or gear flexibility, potentially risking both human safety and robot integrity. For a desired trajectory $q_d(t)$, tracking error e is defined to ensure accurate path following.

Rewriting the 3-DoF robot dynamic equation in (19), we obtain (39) and (40) [3].

$$N(q, \dot{q}) = C(q, \dot{q}) + G(q) + F(\dot{q}) \tag{39}$$

$$\tau = M(q)\ddot{q} + N(q, \dot{q}) + \tau_d \tag{40}$$

In (41), the tracking error is defined as follows:

$$e = q_d - q \tag{41}$$

Based on the equation in (13), where q_d is the desired trajectory and q is the actual trajectory, the controller must optimize the robot's movement from an initial to a final point, ensuring the end-effector moves quickly and safely. Figure 8 illustrates the 3-DoF Scorbot model robot implementing the dynamic equation.

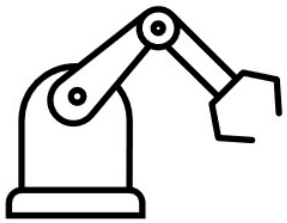


Figure 8. 3-DoF robot.

To demonstrate the influence of the input t on the tracking error, it is derived twice to obtain (42) and (43).

$$\dot{e} = \dot{q}_d - \dot{q} \tag{42}$$

$$\ddot{e} = \ddot{q}_d - \ddot{q} \tag{43}$$

Solving $\ddot{q}$ in (40) and substituting in Equation (43), we obtain (44).

$$\ddot{e} = \ddot{q}_d + M^{-1}(N + \tau_d - \tau) \tag{44}$$

Defining the control input Function (45)

$$u = \ddot{q}_d + M^{-1}(N - \tau) \tag{45}$$

and the disturbance Function (46)

$$\mathrm{w} = M^{-1}\tau_d \tag{46}$$

we obtain (47)

$$x = \begin{bmatrix} e \\ \dot{e} \end{bmatrix} \tag{47}$$

Therefore, the PD Computed Torque Control is defined in Equation (48)

$$\tau = M(q)\left(\ddot{q}_d + K_d\dot{e} + K_p e\right) + N(q, \dot{q}) \tag{48}$$

where

K_p: proportional gain.
K_d: derivative gain.
Figure 9 shows the control scheme that represents Equation (48).

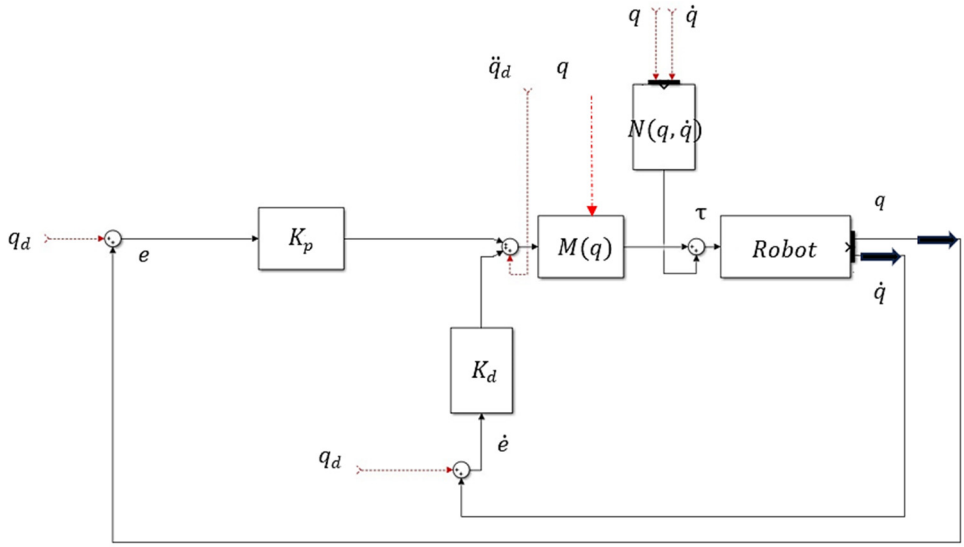


Figure 9. PD Computed Torque Control scheme.

Initially, the parameter values used were selected according to experiments cited in the study articles in order to establish a set of configurations that allowed the controller to function through modification and improvement. One limitation was the complexity of linking the model of the robot designed in Simscape to Simulink for performing 3D animations. The parameter values used with the Computed Torque PD controller were selected from work [1].

For the simulation, the following considerations were established, which are indicated in Table 5.

Table 5. Simulation conditions for PD Computed Torque Controller.

Simulation Conditions	
Gains	kp = 200; Kd = 20
Total simulation time	6 [seg]
External forces	Step function
MATLAB® R2024a	Standard configuration
Trajectory generation	Trapezoidal Velocity with Waypoints
Data Capture	3D Array

Proportional–Integral–Derivative Computed Torque Control

Similarly, the PID Computed Torque Control is defined in (49).

$$\tau = M(q)\left(\ddot{q}_d + K_d\dot{e} + K_p e + K_i\varepsilon\right) + N\left(q,\dot{q}\right) \quad (49)$$

where

K_i: integral gain.

Analysis of the PID Computed Torque scheme reveals its similarity to the PD Computed Torque, with the addition of an integral component. However, its connection configuration at the summation point deviates from the classic PID control form due to its structural characteristics.

Similarly, the parameter values used with the Computed Torque PID controller were selected from work [1]. For the simulation, the following considerations were established, which are indicated in Table 6. Figure 10 shows the control scheme that represents Equation (49).

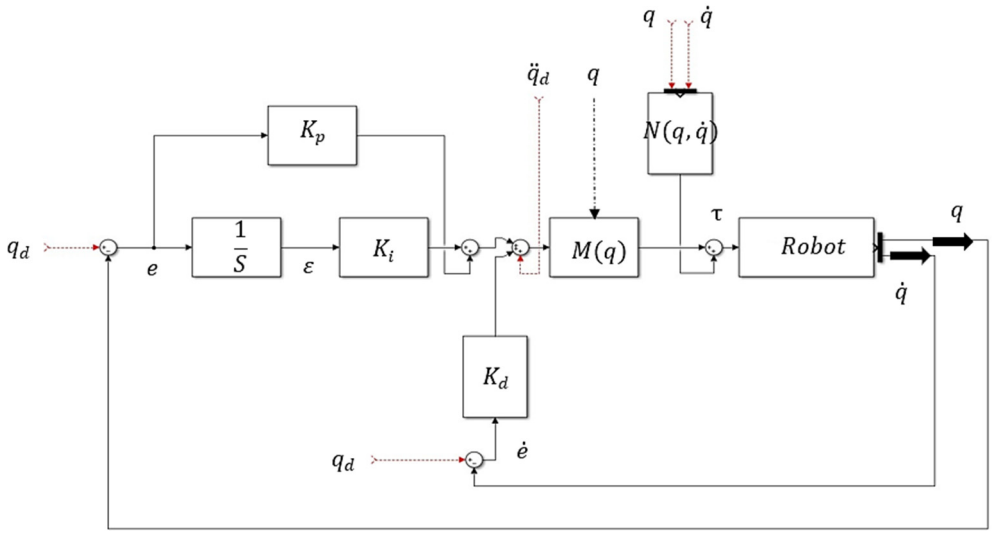


Figure 10. PID Computed Torque Control scheme.

Table 6. Simulation conditions for PID Computed Torque Controller.

Simulation Conditions	
Gains	kp = 200; Ki = 10; Kd = 20
Total simulation time	6 [seg]
External forces	Step function
MATLAB® R2024a	Standard configuration
Trajectory generation	Trapezoidal Velocity with Waypoints
Data Capture	3D Array

4.4.2. Proportional–Derivative–Integral Fuzzy Controller

The fuzzy logic controller is derived from the mathematical logic that analyzes analog inputs whose values are between 0 and 1, unlike digital inputs, which can only be 0 or 1. Figure 11 shows the structure required to implement a fuzzy-type control.

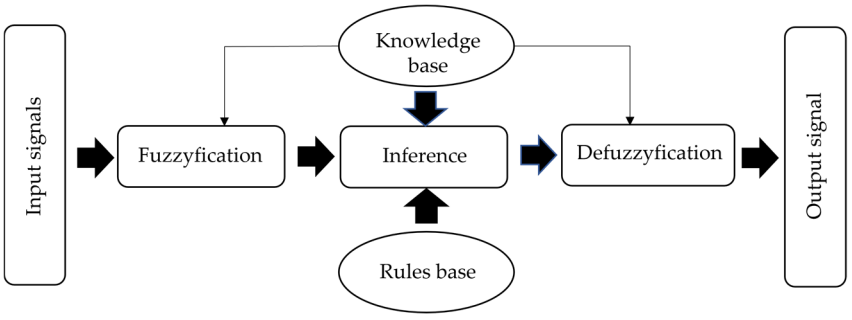


Figure 11. General structure of a fuzzy controller.

Given that classical control methods prove ineffective and unreliable in managing robots subject to various uncertainties—including model variations, payload changes, and parameter fluctuations—the integration of FLC with traditional control methods is proposed to enhance system robustness and effectiveness. The implemented fuzzy controller employs the Mamdani inference method, incorporating nine triangular membership functions for each of the two inputs, e and ė, following the IF-THEN rule. The inputs' linguistic terms are categorized as negative (N), zero (Z), and positive (P). The controller output utilizes five triangular membership functions designated as negative big (NB), negative small (NS), zero (Z), positive small (PS), and positive big (PB). As shown in Figure 12, the fuzzy controller design utilizes the Mamdani configuration with two inputs and one output.

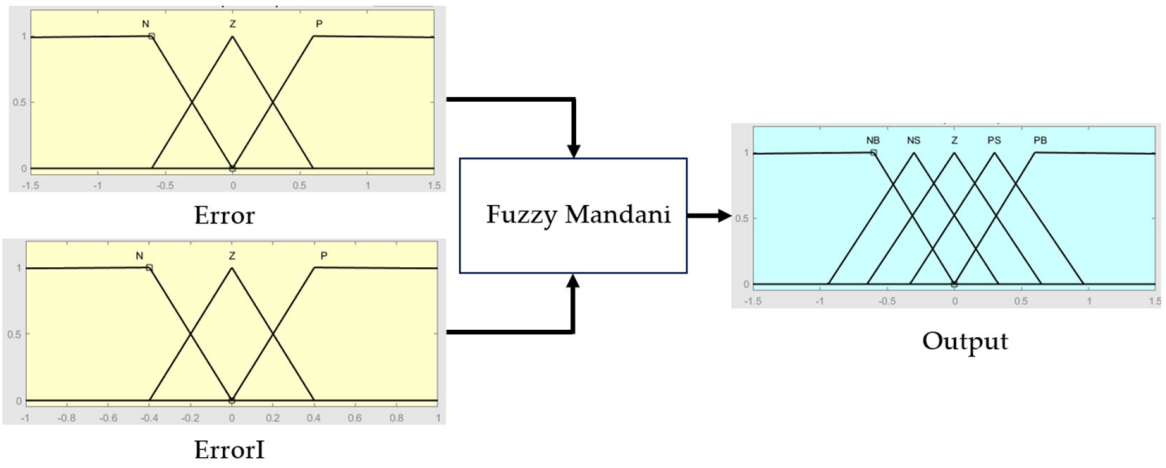


Figure 12. Design of the two-input and one-output fuzzy controller.

Regarding membership conditions, Table 7a,b shows its design.

Table 7. (**a**) Fuzzy logic controller rule. (**b**) Fuzzy logic controller rule expressed based on MacVicar–Whelan [4].

(a)	
Membership Conditions	
IF (Error is P) and (ErrorI is P)	Then (Output is PB)
IF (Error is P) and (ErrorI is N)	Then (Output is Z)
IF (Error is P) and (ErrorI is Z)	Then (Output is PS)
IF (Error is N) and (ErrorI is P)	Then (Output is Z)
IF (Error is N) and (ErrorI is N)	Then (Output is NB)
IF (Error is N) and (ErrorI is Z)	Then (Output is NS)
IF (Error is Z) and (ErrorI is P)	Then (Output is PS)
IF (Error is Z) and (ErrorI is N)	Then (Output is NS)
IF (Error is Z) and (ErrorI is Z)	Then (Output is Z)

(**b**)			
Error/ErrorI	**N**	**Z**	**P**
N	NB	NS	Z
Z	NS	Z	PS
P	Z	PS	PB

For the simulation, the following considerations were established, which are indicated in Table 8.

Table 8. Simulation conditions for PD-PI + fuzzy controller.

Simulation Conditions	
Gains	kp = 20; Ki = 1; Kd = 5; Ku = 20; Ku1 = 20
Total simulation time	6 [seg]
External forces	Step function
MATLAB® R2024a	Standard configuration
Trajectory generation	Trapezoidal Velocity with Waypoints
Data Capture	3D Array

Figure 13 shows the detail of the PD-PI + fuzzy controller.

Likewise, the parameter values used with the PD-PI + fuzzy controller were selected from work [18].

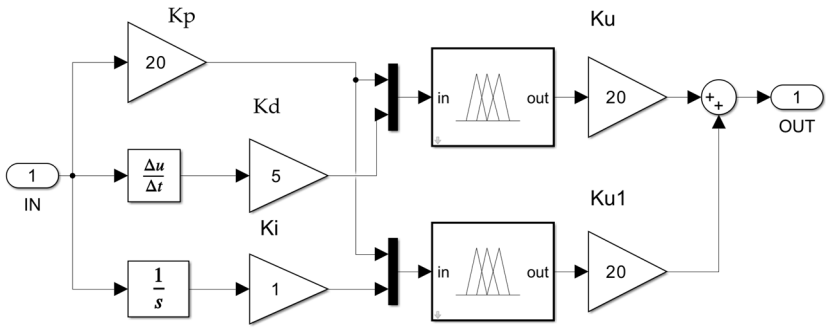


Figure 13. PD-PI + fuzzy control.

4.4.3. Model Predictive Control

The concept of Model Predictive Control encompasses a range of control methods that explicitly utilize a process model to generate control signals through the minimization of an objective function. These methodologies share common structural elements, including the following:

- A predictive model for determining process evolution at future time instants.
- An objective function minimization process.
- Implementation of a finite and sliding control horizon, which involves computing the control sequence across the entire horizon, executing only the first signal, and iterating the process at each sampling instant.

Figure 14 presents the general structure for predictive control implementation. The system employs a model to forecast process output or state evolution based on known input and output signals. An optimizer calculates future control actions while considering the cost function and applicable constraints.

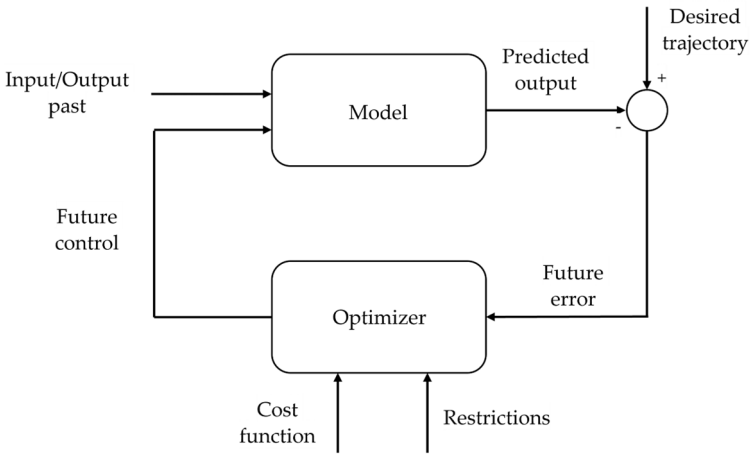


Figure 14. General structure of an MPC.

In the same way, the parameter values used with the MPC controller were selected from [31].

For the simulation, the following considerations were established, which are indicated in Table 9.

Table 9. Simulation conditions for MPC.

Simulation Conditions	
Sample time	0.001 [seg]
Total simulation time	6 [seg]
External forces	Step function

Table 9. *Cont.*

Simulation Conditions	
MATLAB® R2024a	Standard configuration
Trajectory generation	Trapezoidal Velocity with Waypoints
Data Capture	3D Array

Figure 15 shows the details of the MPC.

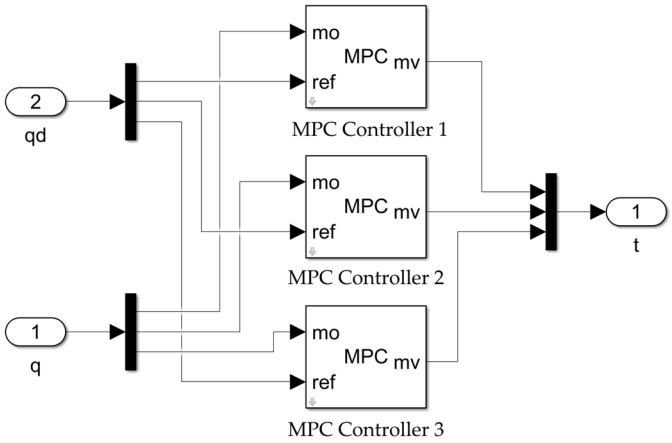


Figure 15. MPC design.

4.5. System Identification

Implementing robot control requires prior knowledge of the desired trajectory for task execution. Trajectory planning presents multiple challenges, including obstacle avoidance and ensuring actuator limitations for voltage and torque are not exceeded. The fundamental control objective is to guide the robot along a desired trajectory defined by a set of end-effector coordinates.

In robotics, tasks are typically specified in the workspace or Cartesian space, where manipulator movements can be naturally described relative to the environment and workpiece. However, trajectory tracking control is more effectively implemented in joint space, which facilitates the formulation of arm dynamics. Consequently, inverse kinematics becomes essential for converting the desired Cartesian trajectory into joint space coordinates $q_d(t)$.

Recent years have seen increased research interest in redundant robot manipulators. Kinematic redundancy occurs when robot manipulators possess more DoF than required for a given task. This characteristic enhances robot dexterity, enabling additional tasks such as obstacle avoidance or singularity prevention while maintaining the primary motion task. For instance, a 3-DoF robot manipulator exhibits functional redundancy when controlling only end-effector position (two coordinates), but not when both position and orientation are required.

4.6. Cartesian Trajectories

The evaluation of the proposed control strategy involves tracking performance analysis of two trajectories. Figure 16 details the designed robot trajectory. The intersection of vertices and straight lines can generate abrupt changes in speed and acceleration, potentially inducing vibrations or undesired movements if not properly managed by the controller. This creates an ideal scenario for assessing controller robustness and adaptability.

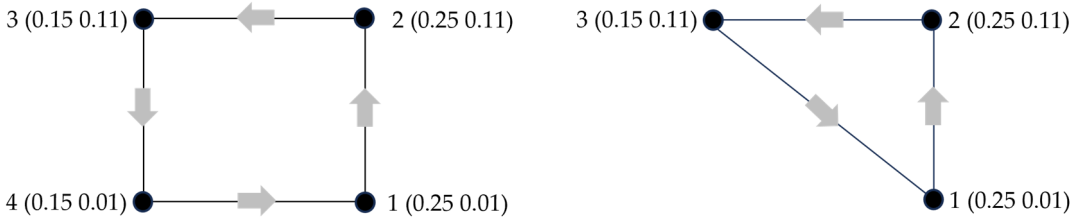


Figure 16. Trajectories in Cartesian space. (**a**) Sequence A. (**b**) Sequence B.

The coordinate points are recorded in a matrix expressed in X, Y, and Z, and time, as shown in Table 10.

Table 10. Coordinate point matrix for sequence A.

		Time [s]					
		1	2	3	4	5	6
Cartesian Axis [m]	X	0.2500	0.2500	0.2500	0.1500	0.1500	0.2500
	Y	0.0100	0.0100	0.1100	0.1100	0.0100	0.0100
	Z	0.1100	0.1100	0.1100	0.1100	0.1100	0.1100

Similarly, Table 11 shows the matrix of X, Y, and Z coordinates and time corresponding to Sequence B.

Table 11. Coordinate point matrix for sequence B.

		Time [s]					
		1	2	3	4	5	6
Cartesian Axis [m]	X	0.2500	0.2500	0.2500	0.1500	0.2500	0.2500
	Y	0.0100	0.0100	0.1100	0.1100	0.0100	0.0100
	Z	0.1100	0.1100	0.1100	0.1100	0.1100	0.1100

Figure 17a,b shows the desired trajectories A and B of the joints over time.

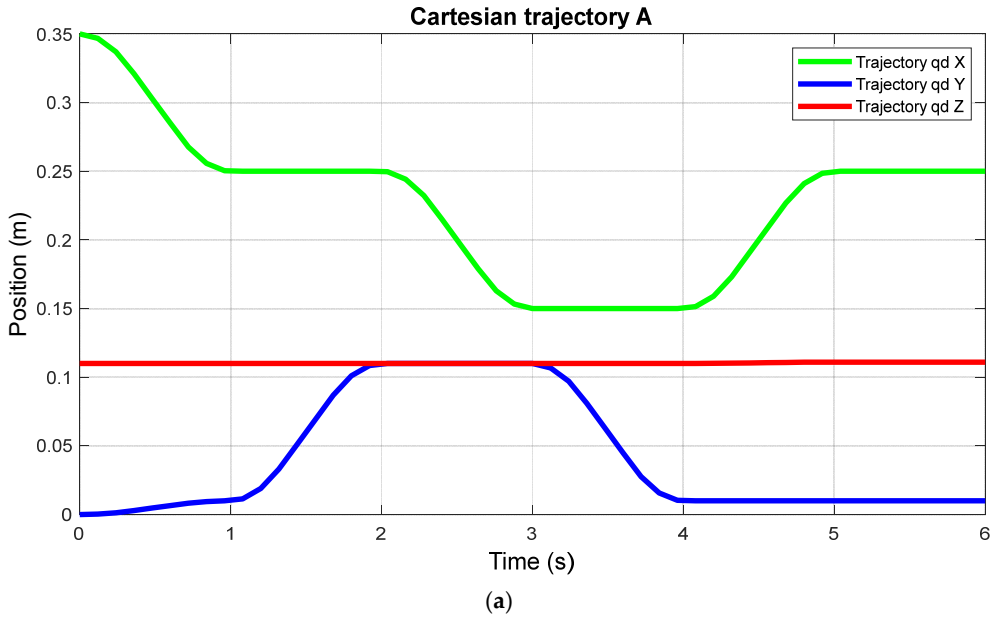
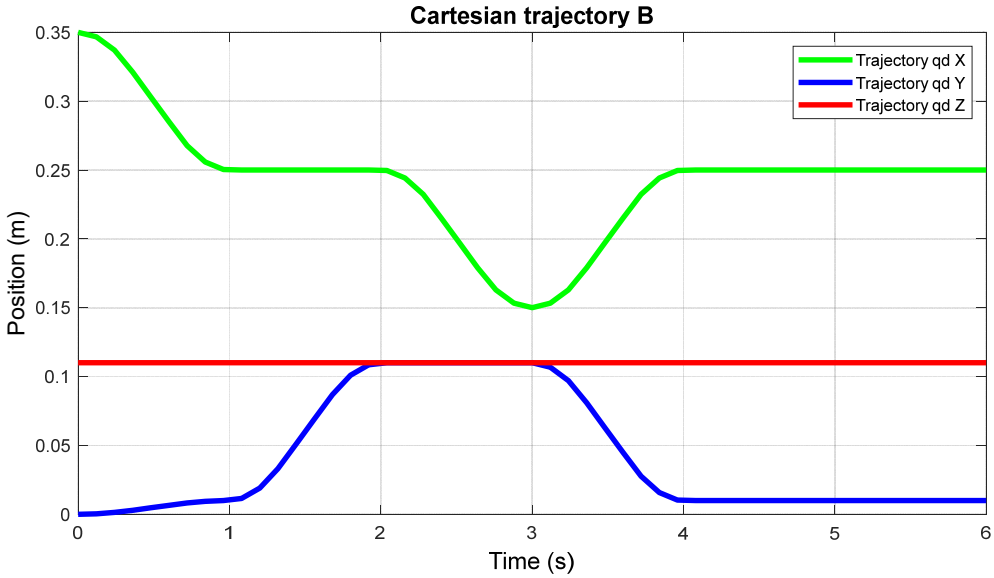


Figure 17. *Cont.*



(**b**)

Figure 17. Trajectories in Cartesian space, as a function of time. (**a**) Sequence A. (**b**) Sequence B.

4.7. Performance Indexes

The quantitative assessment of controller performance in trajectory tracking utilizes specific Performance Indices (PIs). The selected metrics include: Residual Mean Square, Residual Standard Deviation, and Index of Agreement, as defined in Equations (50)–(52).

$$\mathrm{RMS} = \sqrt{\frac{\sum_{i=1}^{n}(o_i - p_i)^2}{n}} \tag{50}$$

$$\mathrm{RSD} = \sqrt{\frac{\sum_{i=1}^{n}(o_i - p_i)^2}{\sum_{i=1}^{n} {o_1}^2}} \tag{51}$$

$$\mathrm{IA} = 1 - \frac{\sum_{i=1}^{n}(o_i - p_i)^2}{\sum_{i=1}^{n}(|o_{i'}| + |p_{i'}|)^2} \tag{52}$$

with

$$p_{i'} = p_i - o_m$$
$$o_{i'} = o_i - o_m$$

where o_i represents the observed value and p_i the predictive value.

5. Results

This section evaluates simulation results from four controllers: Computed Torque PD, Computed Torque PID, PD-PI fuzzy, and MPC, using these three performance indicators. Additionally, the analysis encompasses PID and PID with Gain Scheduling controllers, along with their performance in obstacle avoidance trajectories.

The transient response in the Cartesian space of all controllers has improved with the proposed trajectory design, bringing the desired trajectory points closer to the Cartesian position given by the initial conditions in the 3D plots.

5.1. Computed Torque Proportional–Derivative Control

When simulating trajectory sequence A, Figure 18 shows the desired and actual trajectories in the XYZ Cartesian plane.

The following graphs show trajectories A and B in 3D. Figure 19 illustrates the actual trajectory A in 3D.

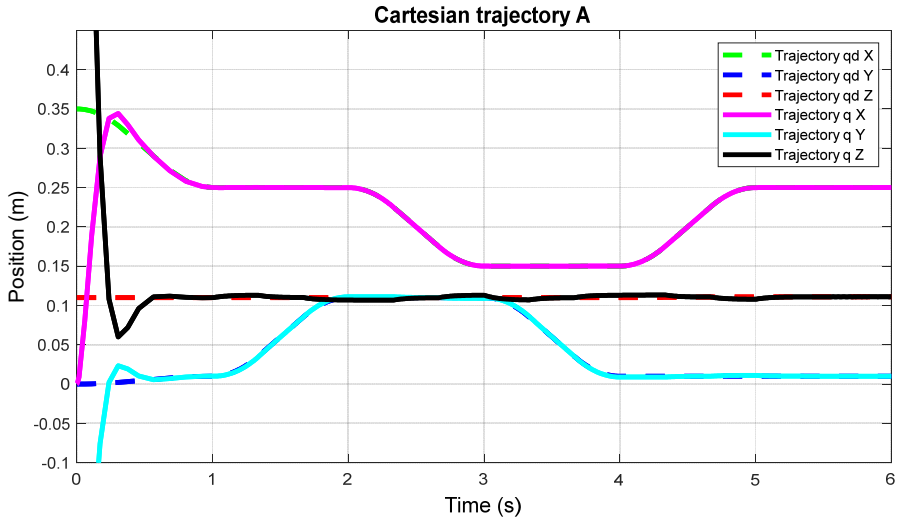


Figure 18. Sequence of trajectory A, as a function of time, using Computed Torque PD control.

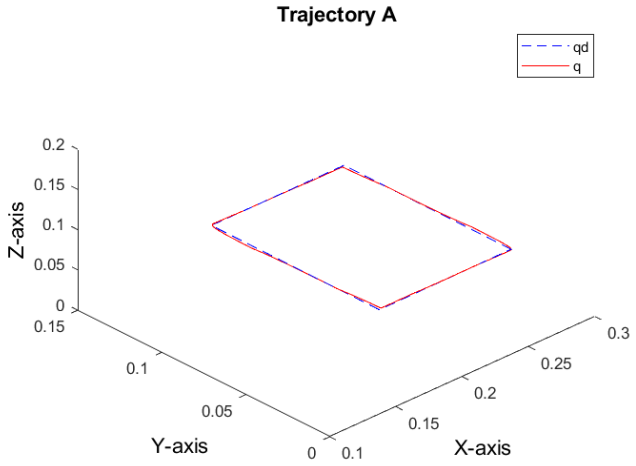


Figure 19. 3D graph of trajectory A.

To confirm the performance of the Computed Torque PD controller, trajectory B is used. Figure 20 shows the graph of the actual trajectory B in 3D.

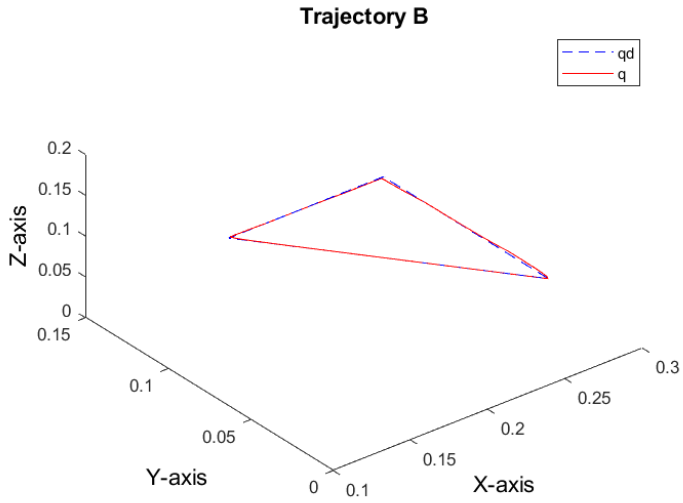


Figure 20. 3D graph of trajectory B.

Analysis of Figures 16–18 demonstrates exceptional trajectory execution by the controller, exhibiting minimal displacement along the XYZ coordinate axes.

5.2. Computed Torque Proportional–Integral–Derivative Control

For trajectory sequence B simulation, Figure 21 illustrates the comparison between desired and actual trajectories in the XYZ Cartesian plane.

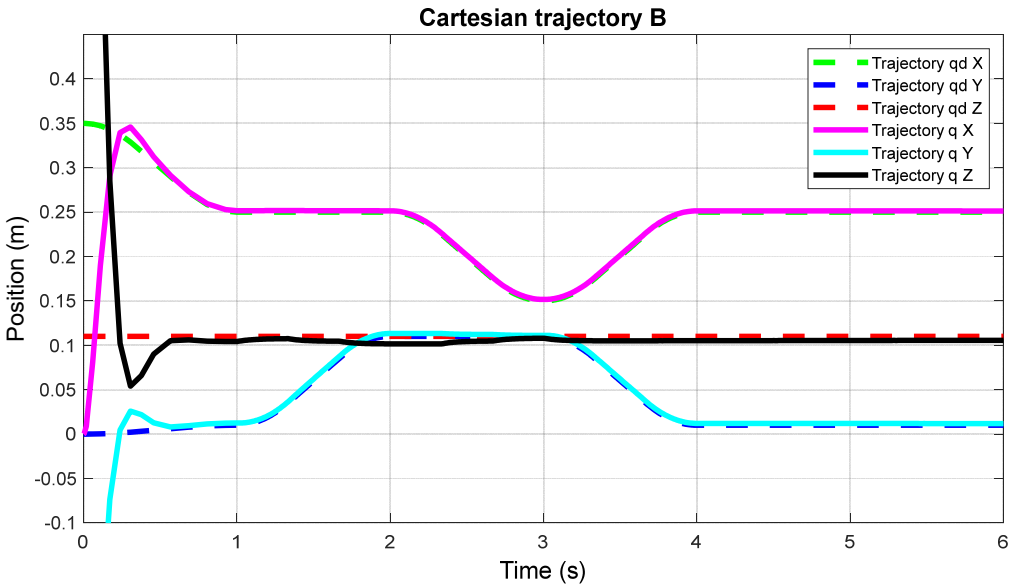


Figure 21. Sequence of trajectory B, as a function of time, using Computed Torque PID control.

Figure 22 illustrates the graph of the actual trajectory in 3D.

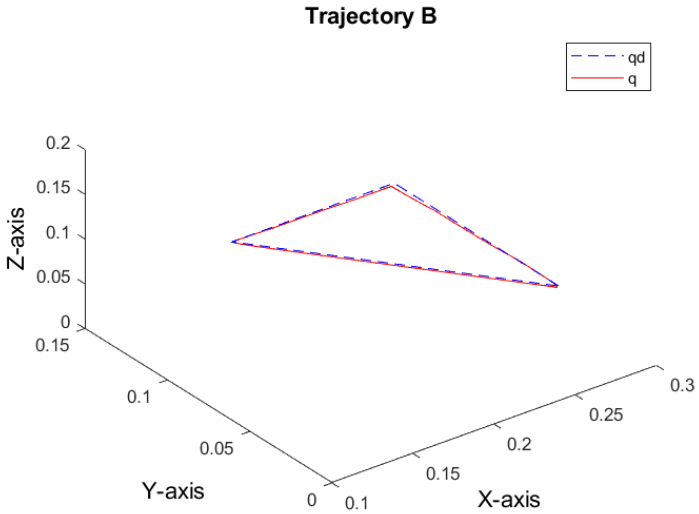


Figure 22. 3D graph of trajectory B.

The results presented in Figures 21 and 22 confirm accurate trajectory execution by the controller, with minimal displacement along the Z coordinate axis.

5.3. Proportional–Derivative–Integral Fuzzy Control

Figure 23 illustrates the comparison between desired (q_d) and actual (q) trajectories for sequence A.

Figure 24 illustrates the graph of the actual trajectory in 3D.

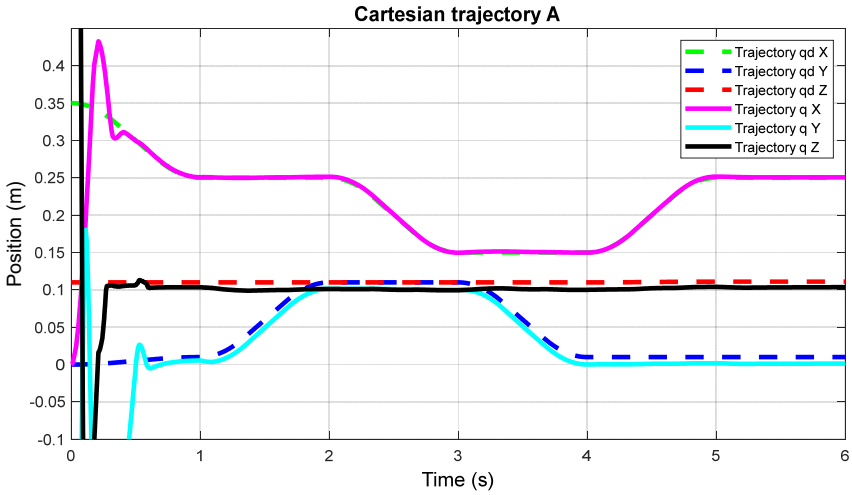


Figure 23. Sequence of trajectory A, as a function of time, using PD-PI fuzzy control.

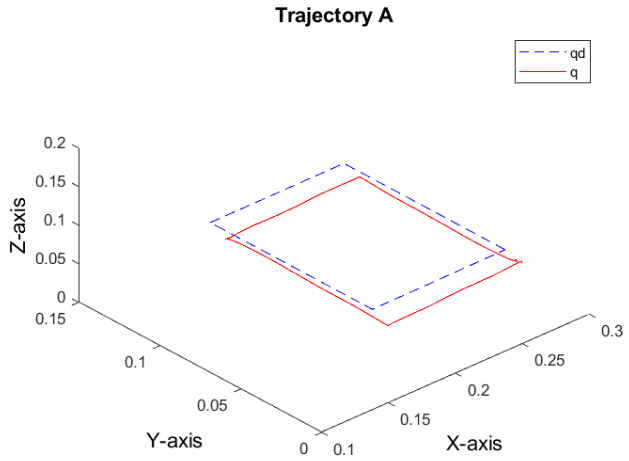


Figure 24. 3D graph of trajectory A.

Figures 23 and 24 demonstrate that the controller successfully tracked the desired trajectory with negligible deviations along the YZ coordinate axes.

5.4. Model Predictive Control

Figure 25 illustrates the trajectory profiles of q_d and q with sequence A.

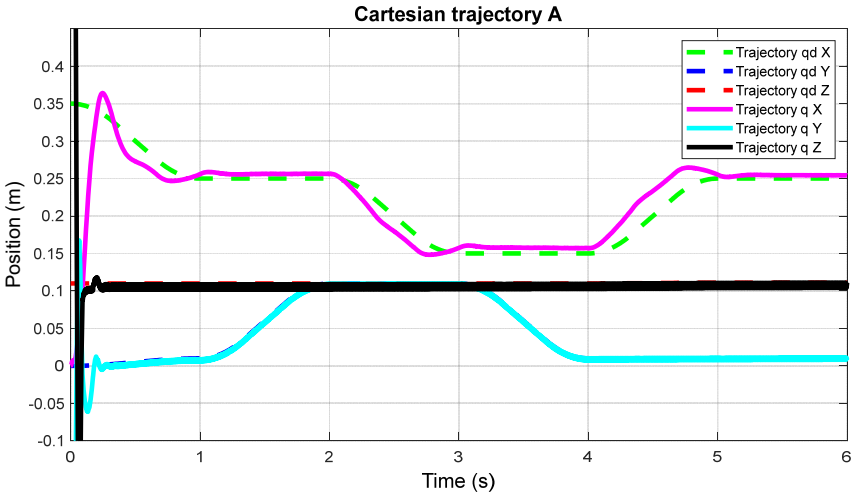


Figure 25. Sequence of trajectory A, as a function of time, using MPC.

Figure 26 shows the graph of the actual trajectory in 3D.

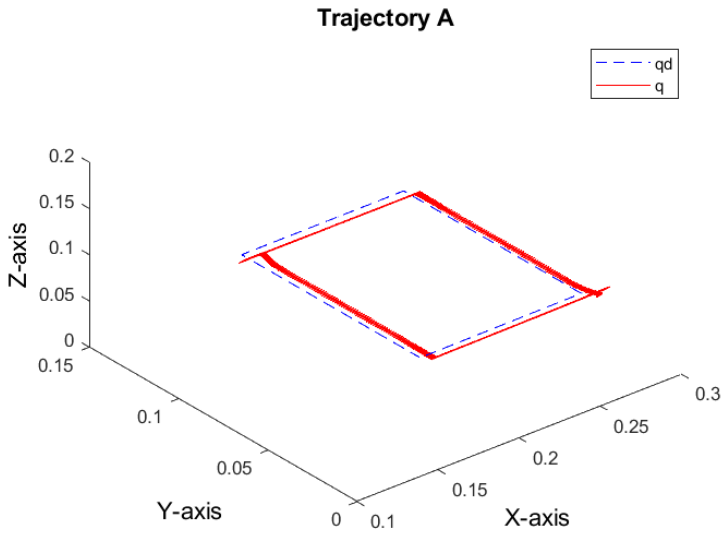


Figure 26. 3D graph of trajectory A.

Figures 25 and 26 show that the controller successfully followed the desired trajectory, with minimal displacement in the XY coordinate axes.

Table 12 presents the performance indices achieved by the three controllers that were analyzed, designed, and simulated.

Table 12. Comparative table of performance indices.

Controller	Computed Torque PD			PD-PI Fuzzy			MPC		
Index	**RMS**	**RSD**	**IA**	**RMS**	**RSD**	**IA**	**RMS**	**RSD**	**IA**
X Coordinate	0.0720	0.1595	0.9940	0.0641	0.0133	0.9989	0.0675	0.0161	0.9981
Y Coordinate	0.0904	0.1731	0.9885	0.0742	0.0451	0.9991	0.0721	0.0572	0.9984
Z Coordinate	0.1447	0.1443	0.9709	0.0563	0.0246	0.9976	0.0981	0.0401	0.9979
Trajectory A					√				

Next, a quantitative comparison and interpretation of the dynamic performance of the designed controllers is presented. It is important to note that the RMS values range from 0.0720 to 0.1447 for the Computed Torque PD controller, indicating precision in tracking the planned trajectory. For the Residual Standard Deviation index, the values range from 0.0161 to 0.0572 for the Model Predictive Control controller, indicating good trajectory tracking, although with vibration in some segments. For the Index of Agreement, the values range from 0.9976 to 0.9991 for the PD-PI fuzzy controller, indicating smoothness and stability of the robot.

When comparing the three control methodologies, the Computed Torque Controllers were the most efficient, under conditions influenced by fixed loads applied to the robot joints. The Computed Torque PD controller stands out for its advantages in terms of accuracy, robustness, simplicity, and applicability in real-world scenarios, such as medicine and industrial agriculture; in the latter case, to perform tasks such as painting, welding, assembly, and manipulation of parts and components, among others.

5.5. Trajectory Tracking

To compare the operation, PID control is simulated, considering predefined trajectories, including “evade from the side” and “evade from above”.

5.6. Classical Proportional–Integral–Derivative Control

To compare the effectiveness of the controllers, PID control is implemented and defined in (53) and (54).

$$\dot{\varepsilon} = e \tag{53}$$

$$\tau = K_p e + K_i \varepsilon + K_v \dot{e} \tag{54}$$

Proportional–Integral–Derivative Control

When simulating the PID controller with trajectory A, the graph in Figure 27 is obtained.

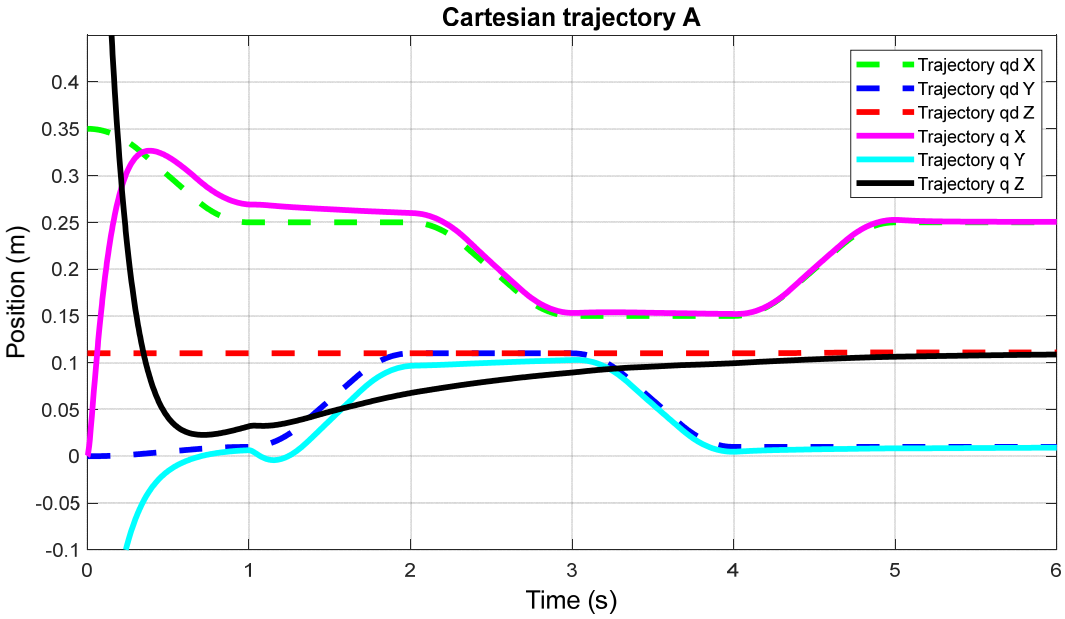


Figure 27. Sequence of trajectory A, as a function of time, using PID control.

Figure 28 shows the graph of the actual trajectory in 3D.

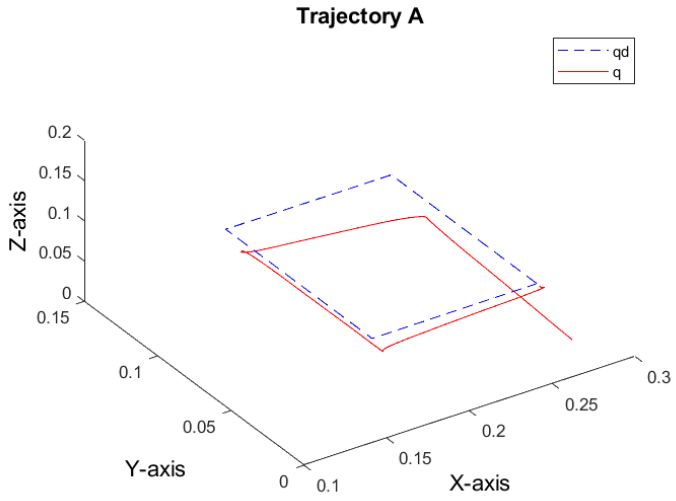


Figure 28. 3D graph of trajectory A.

When simulating trajectory B, Figure 29 shows the graph of the actual trajectory in 3D.

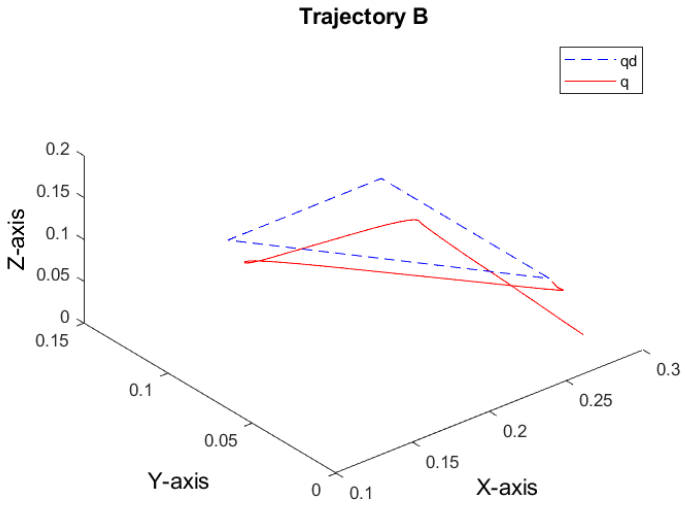


Figure 29. 3D graph of trajectory B.

Figures 27–29 indicate that the controller achieved the desired trajectory, although a notable displacement was observed in the Z coordinate axis.

5.7. Proportional–Integral–Derivative Control with Gain Scheduling

The Gain Scheduling method relies on a table specifying the controller parameters to be used for each DoF. In cases where the operating point does not exactly match the table values, an interpolation technique is recommended [1,3]. The constants K_p, K_i, and K_d are obtained through testing. Figure 30 illustrates the PID control with Gain Scheduling.

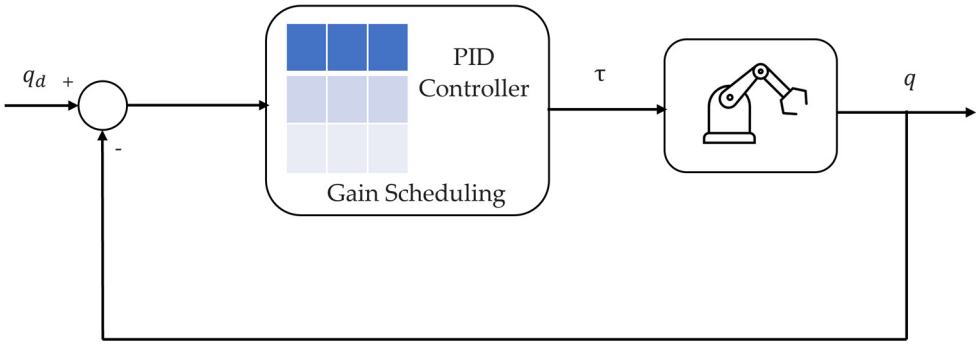


Figure 30. PID control with Gain Scheduling.

Table 13 shows the details of the constants used in the simulation with PID control with Gain Scheduling.

Table 13. Constants for PID Control with Gain Scheduling.

Link	K_p	K_i	K_d
1	1500	100	50
2	2000	200	150
3	3000	300	50

Next, trajectory A is simulated. Figure 31 shows the graph of trajectories q_d and q, while Figure 32 presents the 3D graph.

Figures 31 and 32 indicate that the controller executed the desired trajectory with negligible displacement in the XYZ coordinate axes, showing performance similar to the Computed Torque Controllers.

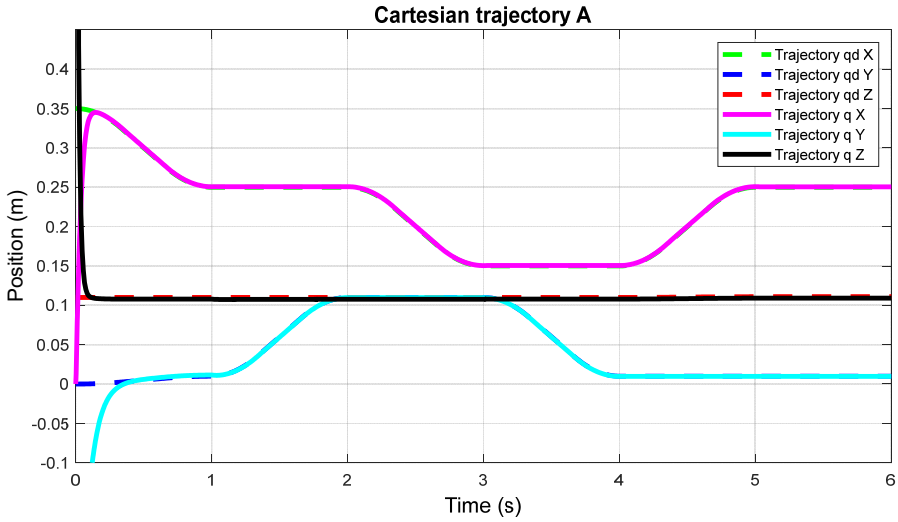


Figure 31. Sequence of trajectory A, as a function of time, using PID control with Gain Scheduling.

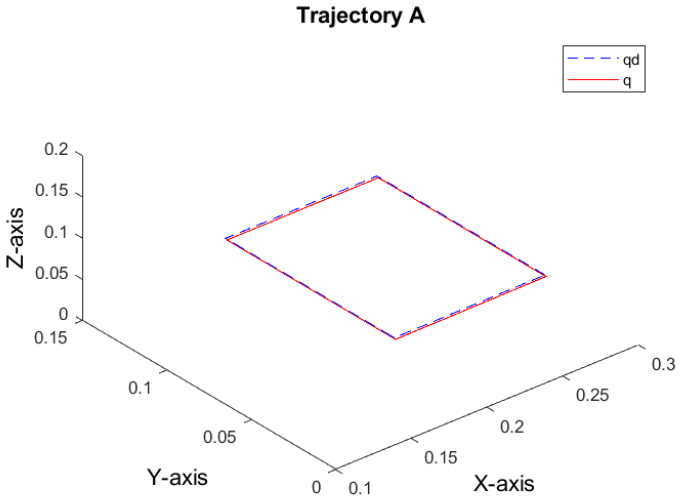


Figure 32. 3D graph of trajectory A.

5.8. "Evade from the Side" Trajectory

Figure 33 shows the 3D graph of the "evade from the side" trajectory.

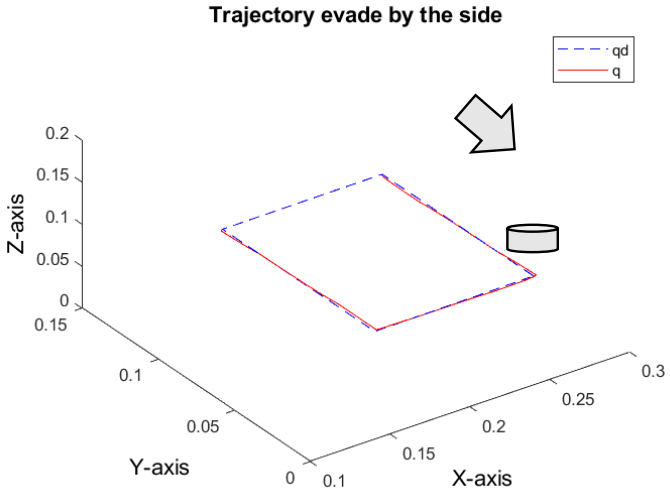


Figure 33. 3D graph of trajectory A, evading from the side.

5.9. "Evade from Above" Trajectory

Figure 34 shows the 3D graph of the "evade from above" trajectory.

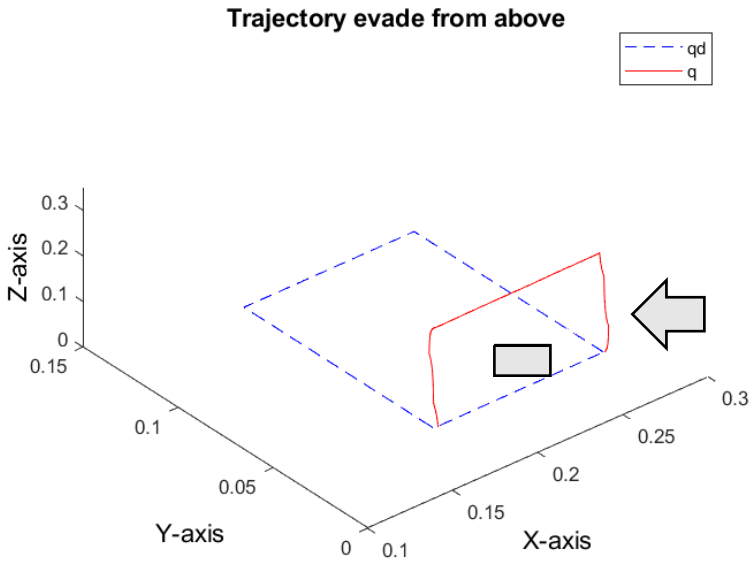


Figure 34. 3D graph of trajectory A, evading from above.

Figures 33 and 34 show that the Computed Torque PD controller successfully followed the desired trajectory with minor vibration in the Z coordinate axis in the "Evade from Above" trajectory. All controllers executed the trajectories acceptably, considering the operation time and spatial movements of the robot's joints.

In general, the controllers performed the desired trajectory adequately; deviations were observed in the *X*-axis with the PD-PI fuzzy controller and in the *Z*-axis for the MPC controller. The PID controller showed the worst response, while the others displayed similar behavior, with the Computed Torque PD, Computed Torque PID, and PID with Gain Scheduling controllers standing out as the most accurate. Visually, the PID controller with Gain Scheduling was the second best, with the Computed Torque PD controller showing the best performance.

6. Discussion

The main objective of this study was to compare the performance of controllers through graphs and performance indices associated with trajectory planning for the robot, how to avoid obstacles, and ensure that the path does not exceed the voltage and torque limitations of the actuators. Computed Torque Control is an effective technique that provided good results, achieving smoothness in movement.

The PI-PD fuzzy and MPC controllers handled abrupt changes at vertices during trajectory tracking acceptably, exhibiting moderate vibrations. All controllers showed good precision in controlling the desired trajectory.

Additionally, a Scorbot robot model was developed in Simscape to visualize its movements through animations and verify its displacement in the XYZ plane using the study trajectories. This work contributes to the visualization of the behavior of real trajectories in the robot's space-time in the presence of external disturbances and abrupt changes according to the nature of the selected trajectories.

In the performance graphs of the three controllers, high precision in trajectory tracking is observed. This is confirmed by their RMS indices, which yield RSDs of approximately 1 cm in the XYZ coordinate axes, while the IAs reach values close to one.

The PID controller remains widely used globally, and as we can see from the obtained results, using it with Gain Scheduling achieves excellent trajectory tracking.

In real-world situations, trajectory planning is carried out to minimize accidents and avoid obstacles. In this regard, the proposed trajectories are presented in three-dimensional graphs, through which it is verified that the performance of the Computed Torque PD controller is excellent.

Table 14 summarizes the weaknesses, strengths, and complementary characteristics of the controllers, extracted from this study.

Table 14. Weaknesses, strengths, and complementary characteristics of the controllers.

Controller	Weaknesses	Strengths	Complementary Characteristics
Computed Torque PD	Requires the dynamic model of the robot.	Simple, provides precision and smoothness in movement.	Robust in selected trajectories. Low computational cost. Negligible processing/execution time.
PD-PI Fuzzy	Needs additional classical control.	Does not require a dynamic model. Provides smoothness.	Acceptable in trajectories. Low computational cost. Significant processing/execution time.
MPC	Requires the dynamic model of the robot.	Predicts future values. Acceptable movement.	Acceptable in trajectories. Low computational cost. Minimal processing/execution time.

Practical Applications

Currently, many robotic systems are used in industry, agriculture, and healthcare. For example, the authors in [29] present the simulation of two control schemes that allow decoupling of the joint control scheme, plus a dynamic compensation loop—a feature that gave it the name PD+ and Computed Torque for manipulator robot trajectory using the dynamic model of the CoBot UR5. Their results indicate that both controllers performed adequately in trajectory tracking.

In precision agriculture, robotic arms present significant technical advantages, such as improved operational precision and efficiency [32]. Today, harvesting heavy crops is considered a laborious task, specifically in Japan, which has faced a severe labor shortage in agricultural fields [33]. The authors present a procedure for the development and evaluation of a 4-DoF articulated robotic arm, designed as a drive unit for an outdoor agricultural environment tractor robot.

In the medical field, the authors in [34] analyze the control of a 7-DoF robotic arm for medical delivery, generating a point cloud map of the workspace of the redundant robotic arm within the allowable joint range to obtain the foundations for motion control and trajectory planning of medical delivery robots. In [35], the authors indicate that robotics and artificial intelligence are gaining increasing importance in the pharmaceutical industry. Future applications of artificial intelligence and robots in healthcare include the development and manipulation of medications, identification of fatal diseases, remote prescription monitoring, and geriatric care, among others.

In [36], the authors report that the pharmaceutical industry is undergoing rapid progress due to advancements in digitalization and automation, from drug invention to administration. Robotics plays a vital role in complex processes such as research, development, production, and packaging. Worker safety, precision, drug delivery speed, and product quality have also improved due to the development of robots. As a result, today, a large volume of production robots is seeking more roles in small to medium-sized menus. Robotics in the pharmaceutical field accelerates the manufacturing process, allowing multiple medications to be prepared in a short time.

In [37], the authors indicate that bolt support is the main form of support for underground roadways in coal mines in China. Currently, there are some problems in domestic coal mines, such as low efficiency and poor safety. The authors designed a structural scheme for an omnidirectional, multi-angle automatic roof bolter. First, based on the DH

method, the kinematic model of the arm structure was established, combined with MATLAB's kinematic model. An experiment was conducted to simulate the spatial position of the drilling arm's end of the robot and anchor drilling. The research shows that the robot can quickly replace the anchor drilling arm and ensure the quality of the integral anchor drilling, thus meeting the requirements for comprehensive support.

In [38], it is indicated that the shortage of human labor is increasing; therefore, it is expected that more machinery and agricultural equipment will enter the sector. One of the agricultural machinery widely studied today involves robotic arms. The ideal structure of the robotic arm with optimal length is gaining popularity and is used in many sectors, such as manufacturing and agriculture. In their study, the authors use the direct kinematic modeling method to design a robotic arm with 4 degrees of freedom optimized to achieve a specific coordinate in a dynamic environment that mimics a boom arm installed on a tractor. Through simulation, they apply optimization algorithms to compare the performance of the Artificial Bee Colony optimization algorithm and the Particle Swarm Optimization (PSO) algorithm, which can achieve optimal length with minimal errors. Their results indicate that the best algorithm, namely the PSO algorithm, is evaluated by calculating the Mean Squared Error (MSE) of 0.00108527, the Root Mean Square Error (RMSE) of 0.01678, the Mean Absolute Error (MAE) of 0.004286081, and the end effector position error (error of 0.080557045), where the best algorithm has the lowest error value.

In [31], they present a vision system suitable for the automatic recognition of rolls of coils in automatic packaging machines, a task often performed by human operators. The authors propose employing a robotic system equipped with a 2D vision sensor and an external unit that allows for the calculation of mathematical algorithms and the standard image processing with low computational resource consumption. Additionally, they aim to relieve operators from this exhausting work while maintaining a high productivity rate. Moreover, they indicate that the procedure presented in this document for the robust detection of coil cores using a vision system can be useful in any industrial process where coils are used and hardware resources are limited.

In [39], they indicate that the robotic arm is a device consisting of a movable chain of links connected by joints, which can perform repetitive operations at speeds and with precision that far exceed human operators. They are used to enhance the quality and efficiency of the manufacturing process in the industry, generally for assembly, painting, welding, and pick-and-place operations of parts, among others. However, one of the most challenging problems in completing these tasks is determining the target location of the end effector of the robotic arm. Based on the visual servoing algorithm, the study uses inverse kinematics to execute the pick-and-place operation. First, an object recognition algorithm is implemented to identify the object to be grasped, and then an obstacle avoidance algorithm is developed. The results of the study show that good system performance has been achieved with the three algorithms: first, the object recognition algorithm; second, the obstacle avoidance algorithm; and finally, the pick-and-place operation based on visual servoing. It concludes that the visual servoing algorithm of the robotic arm is suitable for pick-and-place applications.

In [40], they indicate that their study focuses on designing an automatic system for an automotive company in China that allows for high-precision localization and grasping using a proprietary robotic arm guided by monocular vision, meeting the efficiency and precision requirements of automatic gripping of the robotic arm. Furthermore, they indicate that traditional robotic arms can only move according to a planned trajectory and cannot obtain external information, which limits their range of application. To enhance the flexibility of robotic arms, control technologies utilizing visual information obtained from cameras have been developed in recent decades. The authors propose a nonlinear multiparallel surface calibration method based on an adaptive k-segment master curve algorithm, which improves the efficiency of a traditional camera calibration algorithm in calibration accuracy, providing an intelligent, robust, flexible, low-cost, and non-invasive selection system that allows for the precise detection and location of flat objects randomly placed on a conveyor

belt using a single camera. The technology has been successfully applied to a gripping and welding system for automobile parts with FANUC and ABB robots.

It can be noted that most of the investigated works require previously simulating control strategies for the robotic arm oriented toward industry, medicine, agriculture, pharmaceuticals, mining, and others, in order to obtain a safe and rapid way to evaluate the robot's behavior. Additionally, the possibility of implementing innovative control technologies in an industrial proprietary robot is a case of application with high demand; the use of visual information obtained from cameras to generate efficient and precise trajectories associated with the movements of a robotic arm in the detection and manipulation of objects is highlighted [38,40].

It is evident that all areas rely on automation and robotics, making it essential to invest in research on robots with varying DoF, depending on the application.

7. Conclusions

Based on the graphs and results obtained, the following conclusions are drawn:

- A 3-DoF Scorbot model robot was designed using the Toolbox, Simscape, and Robotics System, applying design information according to technical specifications. Simulation tests were performed to observe the movement of the robot's joints with geometric trajectories, considering four- and five-point coordinated paths.
- Controllers (PD Computed Torque, PID Computed Torque, PD-PI + fuzzy, and MPC) were designed and simulated using the dynamic model of the robot. Graphs of the desired and actual trajectories were generated, verifying the sequences of the proposed trajectories.
- Classical PID control was included to observe the robot's behavior under the same trajectories, as well as a PID control with Gain Scheduling to enhance its performance. Trajectories for object evasion were also designed.
- Performance indices were calculated, highlighting that the IA indicator approximated a value close to one.

8. Future Work

Based on the findings of this research, future studies should focus on the following topics:

- Developing a more precise mathematical model for the robot, including all moments of inertia.
- Exploring new control strategies.
- Implementing visual sensors for object detection.
- Object tracking using visual sensors.
- Conducting experimental validations to verify the effectiveness of the proposed methods.

Author Contributions: Conceptualization, C.U. and V.T.; methodology, C.U. and V.T.; software, C.U. and V.T.; validation, C.U. and V.T.; formal analysis, C.U. and V.T.; investigation, C.U. and V.T.; resources, C.U. and V.T.; data curation, C.U. and V.T.; writing—original draft preparation, C.U. and V.T.; writing—review and editing, C.U. and V.T.; visualization, C.U. and V.T.; supervision, C.U. and J.K.; project administration, C.U. and J.K.; funding acquisition, C.U. and J.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: All data are contained within this article.

Acknowledgments: This work was supported by the Faculty of Engineering of the University of Santiago of Chile, Chile.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

ABC	Artificial Bee Colony
ABCMG	Artificial Bee Colony with Multi-elite Guidance
AI	Artificial Intelligence
CMRR	Common Mode Rejection Ratio
DH	Denavit–Hartenberg
DoF	Degree-of-Freedom
FLC	Fuzzy Logic Controller
FLJC	Fuzzy logic Based Joint Controllers
GB	Giga Byte
GHz	Giga Hertz
GPS	Global Position System
IA	Index of Agreement
IoT	Internet of Things
LB-NMPC	Learning Based Nonlinear Model Predictive Control
KKT	Karush–Kuhn–Tucker
MAE	Mean Absolute Error
MIMO	Multiple Input/Output System
MPC	Model Predictive Control
MSE	Mean Squared Error
NMPC	Nonlinear Predictive Control
NPL	Nonlinear Non-Problem
PD	Proportional–Derivative
PDCT	Proportional–Derivative Computed Torque
PI	Proportional–Integral
PID	Proportional–Integral–Derivative
PIDCT	Proportional–Integral–Derivative Computed Torque
PIs	Performance Indices
PSO	Particle Swarm Optimization algorithm
RAM	Random Access Memory
RMS	Residual Mean Square
RMSE	Root Mean Square Error
RSD	Residual Standard Deviation
SQP	Sequential Quadratic Program
3D	Three Dimensions

References

1. Barrientos, A.; Peñín, L.; Balaguer, C.; Aracil, R. *Fundamentos de Robótica*; McGraw Hill: Antequera, Spain, 1997.
2. Craig, J. *Robótica*, 3rd ed.; Pearson: Ciudad de Mexico, Mexico, 2006.
3. Lewis, F.L.; Dawson, D.M.; Abdallah, C.T. *Robot Manipulator Control Theory and Practice Second Edition, Revised and Expanded*; Marcel Dekker, Inc.: New York, NY, USA; Basel, Switzerland, 2004.
4. Matute, A.; Bernal, W. *Fuzzy Logic Techniques in Control Engineering*; Revista Ciencia, Innovación y Tecnología (RCIYT): Buenos Aires, Argentina, 2017; Volume III.
5. Gregori, V.; Morillas, S.; Sapena, A. Examples of fuzzy metrics and applications. *Fuzzy Sets Syst.* **2011**, *170*, 95–111. [CrossRef]
6. Fehr, J.; Schmid, P.; Schneider, G.; Eberhard, P. Modeling, Simulation, and Vision-/MPC-Based Control of a PowerCube Serial Robot. *Appl. Sci.* **2020**, *10*, 7270. [CrossRef]
7. Calli, B.; Dollar, A.M. Vision-Based Model Predictive Control for Within-Hand Precision Manipulation with Underactuated Grippers. In Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA), Singapore, 29 May–3 June 2017.
8. Ostafew, C.J.; Schoellig, A.P.; Barfoot, T.D.; Collier, J. Learning-based Nonlinear Model Predictive Control to Improve Vision-based Mobile Robot Path Tracking. *J. Field Robot.* **2016**, *33*, 133–152. [CrossRef]
9. Abad, A.C.; Ligutan, D.D.; Dadios, E.P.; Cruz, L.J.S.; Del Rosario, M.C.D.; Kudhal, J.N.S. Fuzzy Logic-Controlled 6-DOF Robotic Arm Color-based Sorter with Machine Vision Feedback. *Int. J. Adv. Comput. Sci. Appl.* **2018**, *9*, 21–31. [CrossRef]
10. Roshanianfard, A.; Mengmeng, D.; Nematzadeh, S. A 4-dof scara robotic arm for various farm applications: Designing, kinematic modelling, and parameterization. *Acta Technol. Agric.* **2021**, *24*, 61–66. [CrossRef]
11. Iqbal, J. Modern Control Laws for an Articulated Robotic Arm: Modeling and Simulation. *Eng. Technol. Appl. Sci. Res.* **2019**, *9*, 4057–4061. [CrossRef]

12. Urrea, C.; Saa, D. Design, Simulation, Implementation, and Comparison of Advanced Control Strategies Applied to a 6-DoF Planar Robot, Electrical Engineering Department, Faculty of Engineering. *Symmetry* **2023**, *15*, 1070. [CrossRef]
13. Tay, S.H.; Choong, W.H.; Yoong, H.P. A Review of SCARA Robot Control System. In Proceedings of the 2022 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET), Kota Kinabalu, Malaysia, 13–15 September 2022.
14. Yime, E.; Roldán, J.; Villa, J. Computed torque control of a 2-RR planar parallel robot. *Prospectiva* **2017**, *15*, 85–95. [CrossRef]
15. Soto, I.; Campa, R. Two-Loop Control of Redundant Manipulators: Analysis and Experiments on a 3-DOF Planar Arm. *Int. J. Adv. Robot. Syst.* **2013**, *10*, 85. [CrossRef]
16. Oktarina, Y.; Septiarini, F.; Dewi, T.; Risma, P.; Nawawi, M. Fuzzy-PID Controller Design of 4 DOF Industrial Arm Robot Manipulator. *Comput. Eng. Appl. J.* **2019**, *8*, 123–136. [CrossRef]
17. Mirzadeh, M.; Haghighi, M.; Khezri, S.; Mahmoodi, J.; Karbasi, H. Design Adaptive Fuzzy Inference Controller for Robot Arm. *Int. J. Inf. Technol. Comput. Sci.* **2014**, *6*, 66–73. [CrossRef]
18. Ali, A.M.A.; Sabah, I.A.; Ahmed, A.; Mohammed, A.M.A.A. Modelling and Control Two-Link Robot Arm Based on Fuzzy Logic. *IOSR J. Electr. Electron. Eng.* **2022**, *17*, 41–49.00E1.
19. Camacho, E.F.; Bordons, C. Control Predictivo: Pasado, Presente y Futuro. *Rev. Iberoam. Automática E Informática Ind.* **2004**, *1*, 5–28. Available online: https://polipapers.upv.es/index.php/RIAI/article/view/8031 (accessed on 1 October 2024).
20. Guechi, E.-H.; Bouzoualegh, S.; Zennir, Y.; Blažič, S. MPC Control and LQ Optimal Control of a Two-Link Robot Arm: A Comparative Study. *Machines* **2018**, *6*, 37. [CrossRef]
21. AlAttar, A.; Chappell, D.; Kormushev, P. Kinematic-Model-Free Predictive Control for Robotic Manipulator Target Reaching with Obstacle Avoidance. *Front. Robot. AI* **2022**, *9*, 809114. [CrossRef]
22. Azeez, M.I.; Abdelhaleem, A.M.M.; Elnaggar, S.; Moustafa, K.A.F.; Atia, K.R. Optimization of PID trajectory tracking controller for a 3-DOF robotic manipulator using enhanced Artificial Bee Colony algorithm. *Sci. Rep.* **2023**, *13*, 11164. [CrossRef]
23. Kern, J.; Urrea, C.; Torres, H. Implementation and Comparison of Controllers for Planar Robots. *J. Electr. Eng. Technol.* **2017**, *12*, 926–936. [CrossRef]
24. Kern, J.; Urrea, C.; Verdejo, H.; Agramonte, R.; Becker, C. Trajectory Tracking and Disturbance Rejection Performance Analysis of Classical and Advanced Controllers for a SCORBOT Robot. *Robotics* **2024**, *13*, 48. [CrossRef]
25. MatLab Simulink. 2024. Available online: https://la.mathworks.com (accessed on 1 October 2024).
26. Intelitek Inc. SCORBOT-ER 5Plus User Manual, Catalog # 100016 Rev. C, February 1996. Available online: www.intelitek.com (accessed on 19 July 2024).
27. Martínez, J.; Guerrero, G.; Gerardo, L. Modelado del robot SCORBOT ER-V+. In Proceedings of the Congreso Nacional de Control Automático, UNAM, DF, Mexico, 18–20 October 2006.
28. Mañay, E.D.; Chiliquinga, M.D.; Lza, J.E.I.; Ugsha, H.E.Y.; Castillo, P.S. Diseño por simulación de un control Fuzzy y MPC para un proceso de nivel. *Cienc. Lat. Rev. Científica Multidiscip.* **2022**, *6*, 1951–1970. [CrossRef]
29. Toxqui, B.; Palomino, A.; Martínez, C. Estudio comparativo de los esquemas de control Par Calculado y PD+ para el robot UR5, México, Tecnológico Nacional de México en Celaya. *Pist. Educ.* **2023**, *44*, 826–840.
30. Abdo, B.; Gorashi, M. Modeling and Control of 5 DOFs Robotic Arm Using PID and Fuzzy Logic Controller. *Res. Propos.* **2024**, *1*, 1–85. [CrossRef]
31. Comari, S.; Carricato, M. Vision-Based Robotic Grasping of Reels for Automatic Packaging Machines. *Appl. Sci.* **2022**, *12*, 7835. [CrossRef]
32. Jin, T.; Han, X. Robotic arms in precision agriculture: A comprehensive review of the technologies, applications, challenges, and future prospects. *Comput. Electron. Agric.* **2024**, *221*, 108938. [CrossRef]
33. Roshanianfard, A.; Noguchi, N.; Kamata, T. Design and performance of a robotic arm for farm use. *Int. J. Agric. Biol. Eng.* **2019**, *12*, 146–158. [CrossRef]
34. Wang, W.; Wang, P.; Zhao, Y.; Zhu, Y.; Zhao, J. Kinematics and workspace analysis of a robotic arm for medical delivery robots. *InE3S Web Conf.* **2021**, *257*, 01067. [CrossRef]
35. Khandagale, S.; Bappasaheb, N.; Khose, A.; Kshirsagar, P.; Nemane, P.; Haribhau, R. Role of Pharmaceutical Automation and Robotics in Pharmaceutical Industry: A Re-view. *Syst. Rev. Pharm.* **2024**, *15*, 131–135. Available online: https://www.sysrevpharm.org/articles/role-of-pharmaceutical-automation-and-robotics-in-pharmaceutical-industrya-short-review.pdf (accessed on 1 October 2024).
36. Nandi, S.; Das, S. Robotics in the Pharmaceutical Industry. *Hum. J.* **2021**, *21*, 325–334.
37. Jun, Z.; Jiguo, W.; Qianyi, Z.; Ning, W.; Chaofan, W.; Lei, W. Design and Workspace Analysis of Drilling Arm of Mining Anchor Drilling Robot. *J. Phys. Conf. Ser.* **2022**, *2202*, 012057. [CrossRef]
38. Zakey, N.E.N.A.; Zaman, M.H.M.; Ibrahim, M.F. Structural Optimization of 4-DOF Agricultural Robot Arm. *J. Kejuruter.* **2024**, *36*, 1127–1134. [CrossRef]
39. Wei, C.Z.; Zaman, M.H.M.; Ibrahim, M.F. Visual Servo Algorithm of Robot Arm for Pick and Place Application. *J. Kejuruter.* **2024**, *36*, 891–898. [CrossRef]
40. Zheng, Z.; Ma, Y.; Zheng, H.; Gu, Y.; Lin, M. Industrial part localization and grasping using a robotic arm guided by 2D monocular vision. *Ind. Robot. Int. J. Robot. Res. Appl.* **2018**, *45*, 794–804. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.